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Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka| Bengaluru – 560064



STUDENT INNOVATION: SWADESHI FOR ATMANIRHAR BHARAT-TRAVEL & TOURISM

AN PROJECT REPORT

Submitted by

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BACHELOR OF TECHNOLOGY

IN

**INFORMATION SCIENCE AND ENGINEERING
(AI AND ROBOTICS)**



PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **Student Innovation: Swadeshi for Atmanirbhar Bharat - Travel & Tourism** is a bonafide work of TANUSHREE M S (20221ISE0021), DARSHAN R H (20221ISE0036) and MANASA L NAIK (20221ISE0010), who has successfully carried out the Project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE AND ENGINEERING (AI AND ROBOTICS) during 2025-2026.

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DECLARATION

We, **TANUSHREE M S, DARSHAN R H, MANASA L NAIK**, final-year B.Tech students in information Science Engineering at Presidency University, Bengaluru, hereby declare that the project titled **STUDENT INNOVATION: SWADESHI FOR ATMANIRHAR BHARAT-TRAVEL & TOURISM** has been carried out by me independently under the prescribed guidelines.

This project is submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Information Science and Engineering** during the academic year 2025–2026.

We further declare that the work presented in this project report is original and has not been submitted, either in part or in full, for the award of any Degree or Diploma to any other University or Institution.

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ABSTRACT

India is a land of unparalleled geographic and cultural diversity, yet the logistical complexity of planning an efficient multi-city trip remains a significant barrier for many domestic and international travelers. While modern travel portals excel at individual hotel or flight bookings, they lack the “intelligent core” required to sequence multiple destinations into a logical, time-efficient, and culturally sensitive itinerary. This project introduces SwadeshiYatra, a bespoke AI-powered travel planning platform designed to address the unique nuances of the Indian tourism landscape.

At the heart of SwadeshiYatra lies a sophisticated Genetic Algorithm (GA) optimization engine. Unlike standard travel tools, our system treats travel planning as a multi-objective optimization problem, evaluating thousands of possible permutations of a traveler’s desired destinations. The algorithm considers a wide array of “Swadeshi” constraints—ranging from specific temple opening and closing hours to seasonal weather disruptions and regional transport connectivity in tier-2 and tier-3 cities. By doing so, it generates a visiting sequence that minimizes backtracking and unnecessary travel time.

The platform is built on a modular, three-tier software architecture, utilizing Python and Flask for the backend engine and React.js for a responsive, user-friendly frontend. Beyond mere route sequencing, SwadeshiYatra functions as a holistic travel companion by recommending local hospitality providers, including budget-friendly hotels and local guides, thereby spotlighting small-scale businesses often overlooked by global giants.

Performance evaluation conducted across various Indian city clusters demonstrates that the GA engine consistently produces itineraries that are within 8.4% of the theoretical mathematical minimum for travel time, achieving this in an average of just 6.2 seconds. By promoting efficient domestic travel and empowering local service providers, SwadeshiYatra serves as a direct contribution to the Atmanirbhar Bharat initiative, proving that indigenous AI solutions can solve local challenges with global-standard efficiency. Future enhancements will focus on real-time API integration for live transport tracking and natural language processing for more conversational user interactions

TABLE OF CONTENTS

Sl. No.	Title	Page No.
	Abstract	i
	List of Figures	iv
	List of Tables	v
	Abbreviations	vi
1.	Introduction	1
	1.1 Background	1
	1.2 Statistics of project	1
	1.3 Prior existing technologies	2
	1.4 Proposed approach	3
	1.5 Objectives	4
	1.6 SDGs	6
	1.7 Overview of project report	8
2.	Literature Review	10
3.	Methodology	19
4.	Project Management	22
	4.1 project timeline	22
	4.2 Risk analysis	23
	4.3 project budget	24
5.	Analysis and Design	26
	5.1 Requirements	26
	5.2 Block Diagram	26
	5.3 System Flow Chart	29
	5.4 Choosing devices	31
	5.5 Designing units	32
	5.6 Standards	32
	5.7 Mapping Deployment Level with Functional Blocks	33
	5.8 Domain model specification	33

	5.9 Communication model	34
	5.10 IoT deployment level	34
	5.11 Functional view	34
	5.12 Mapping IoT deployment level with functional view	35
	5.13 Operational view	35
	5.14 Other Design	35
6.	Software and Simulation	36
	6.1 Software development tools	36
	6.2 Simulation	36
7.	Evaluation and Results	38
	7.1 Test points	38
	7.2 Test plan	39
	7.3 Test result	39
	7.4 Insights	40
8.	Social, Legal, Ethical, Sustainability and Safety Aspects	41
	8.1 Social aspects	41
	8.2 Legal aspects	41
	8.3 Ethical aspects	42
	8.4 Sustainability aspects	42
	8.5 Safety aspects	43
9.	Conclusion	44
	References	46
	Appendix A	48
	Appendix B: Publication Details	57

LIST OF FIGURES

Sl. No.	Figure Name	Caption	Page No.
1	Figure 1.1	Sustainable development goals	7
2	Figure 3.1	Agile workflow for project stages	21
3	Figure 5.1	Functional Block Diagram	27
4	Figure 5.2	System flow chart	29
5	Figure 5	Role-Based Login Panel for SwadeshiYatra	48
6	Figure 6	Admin Dashboard Overview Panel	48
7	Figure 7	Guide Registration & Verification Panel	49
8	Figure 8	Guide Application Rejection & Re-Submission Panel	49
9	Figure 9	SwadeshiYatra Home & Planning Interface	50
10	Figure 10	AI-Powered City Travel Planner Interface	50
11	Figure 11	Top Attractions Display Panel (City Guide)	51
12	Figure 12	Journey Planning Input & Customization Panel	52
13	Figure 13	AI-Generated Complete Route Plan Interface	53
14	Figure 14	Budget Analysis & Travel Optimization Panel	54
15	Figure 15	Route Visualization & Navigation Interface	55
16	Figure 16	Detailed Itinerary Stop & Cost Summary Panel	56

LIST OF TABLES

Sl. No.	Table Name	Table Caption	Page No.
1	Table 2.1	Summary of Literature reviews	17
2	Table 4.1	Gantt Chart Representation of Project Planning Timeline	22
3	Table 4.2	PESTEL Analysis for SwadeshiYatra Federated Ecosystem	23
4	Table 7.1	System Evaluation Results	39

ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface
UX	User Experience
JSON	JavaScript Object Notation
LLM	Large Language Model
NLP	Natural Language Processing
TSP	Traveling Salesman Problem
OX	Order Crossover
PBX	Position-Based Crossover
PMX	Partially Mapped Crossover
IoT	Internet of Things
IoTWF	Internet of Things World Forum
RBAC	Role-Based Access Control
JWT	JSON Web Token
DPDPA	Digital Personal Data Protection Act
SDG	Sustainable Development Goal
CAGR	Compound Annual Growth Rate
IDE	Integrated Development Environment
DBMS	Database Management System
SQL	Structured Query Language

Chapter 1

INTRODUCTION

1.1 Background

India is one of the most culturally and geographically diverse countries in the world, offering an extraordinary range of travel experiences — from the snow-capped peaks of the Himalayas to the sun-soaked beaches of Kerala, from the ancient temples of Tamil Nadu to the bustling bazaars of Rajasthan. Yet, despite this incredible richness, a large number of domestic and international tourists struggle with a problem that might seem surprising: too many choices and too little guidance. This phenomenon, widely studied in behavioral psychology, is known as the "Paradox of Choice" — where an abundance of options leads to confusion, poor decision-making, and ultimately, a diminished travel experience.

Today's travel planning landscape is fragmented. A tourist planning a trip to India must simultaneously juggle multiple booking platforms for hotels, separate apps for transport, travel blogs for cultural context, and social media for recommendations — all of which give different, often conflicting information. Existing platforms like MakeMyTrip, Yatra, and Google Trips provide static listings and basic booking facilities, but none of them truly understand the traveler as an individual. They do not account for personal preferences, group dynamics, dietary needs, or budget constraints in a holistic way.

SwadeshiYatra was born out of this gap. It is an AI-native travel ecosystem designed specifically for Indian tourism that goes far beyond being a simple booking site. It acts as a Digital Concierge — one that understands user psychology, handles complex multi-city logistics, and connects travelers only to verified, quality-checked service providers. The platform bridges the gap between raw information and actionable, personalized travel plans, making the experience of discovering India more accessible, efficient, and enriching for every kind of traveler.

1.2 Statistics of project

The scale of Indian tourism makes this problem both urgent and impactful to solve. According to the Ministry of Tourism, India received approximately 9.24 million foreign tourist arrivals

in 2023, while domestic tourist visits crossed 2,300 million — a number that reflects the sheer magnitude of internal travel movement across the country. The Indian travel and tourism market is valued at over USD 199 billion and is projected to reach USD 512 billion by 2028, growing at a compound annual growth rate (CAGR) of approximately 13%.

Despite this growth, digital adoption in travel planning remains uneven. While over 820 million Indians now access the internet, a significant portion of them — particularly first-time travelers and those from Tier-2 and Tier-3 cities — do not have access to consolidated, reliable, and personalized travel planning tools. Studies on tourism user behavior show that travelers spend an average of 38 days across 380 touchpoints before finalizing a trip, highlighting the enormous decision fatigue involved in the process.

Furthermore, India's rich cultural diversity makes generic, one-size-fits-all solutions deeply inadequate. A traveler visiting Varanasi for a spiritual retreat has fundamentally different needs from a group of friends doing a backpacking circuit through Himachal Pradesh. SwadeshiYatra's AI-driven personalization engine is specifically designed to address this nuance — making it not just a product, but a need of the hour for Indian tourism.

1.3 Prior existing technologies

Several commercial and open-source platforms currently exist in the travel technology space, but each of them carries notable limitations when applied to the Indian tourism context.

MakeMyTrip and Yatra are the most popular Indian booking platforms. They excel at transactional tasks — booking flights, hotels, and holiday packages — but function purely as aggregators. They do not generate dynamic itineraries, do not offer cultural context, and do not adapt plans based on group type or personal preferences.

Google Trips (now integrated into Google Travel) uses past booking data and location history to suggest places. While powerful in urban settings, it lacks deep cultural knowledge of Indian heritage sites, does not support multi-day route optimization for complex circuits, and has no mechanism for verifying local service providers such as tour guides.

TripAdvisor offers user reviews and ratings but is heavily dependent on crowd-sourced data, which can be manipulated or outdated. It has no AI planning capability and requires users to

manually piece together their own itineraries from scattered reviews.

Airbnb Experiences and similar platforms connect travelers to local experiences but are limited in geographic coverage for rural and semi-urban Indian destinations, and they do not integrate transport, accommodation, and sightseeing into a single unified plan.

The common thread across all these platforms is their inability to synthesize multiple constraints — budget, time, group dynamics, cultural preferences, and route efficiency — into a single, coherent, executable travel plan. SwadeshiYatra addresses all of these gaps through its AI-driven dual planning system and multi-role administrative ecosystem.

1.4 Proposed approach

SwadeshiYatra takes a fundamentally different approach from existing platforms by treating travel planning as an intelligent problem-solving exercise rather than a simple search-and-book transaction.

The platform is structured around two distinct planning philosophies to cater to different traveler needs:

- **The City Planner** is designed for users who already have a destination in mind and want an exhaustive, localized guide. Using a "Deep Fetch" logic, it combines real-time data from the Google Places API (for current ratings and photos) with Wikipedia (for cultural and historical depth) to generate rich, detailed city profiles. Each output includes historical narratives, best time to visit, transport information, attraction cards with visiting hours and entry fees, and AI-generated cultural insider tips.
- **The Journey Planner** is designed for complex, multi-city travel. It operates through a three-step pipeline: first, users select cities and pick must-visit spots from an AI-generated discovery cloud; second, they calibrate their persona by defining group type, activity level, and dietary preferences; and third, the AI synthesizes all constraints to generate a dynamic, day-wise itinerary. This includes route reordering for travel efficiency, airport suggestions based on the user's home country, and a real-time Feasibility Verdict that checks whether the planned trip fits within the stated budget.

The platform also features a Multi-Role Administrative Ecosystem involving tourists, administrators, and service providers — including hotels, restaurants, and tour guides — each

with their own dedicated dashboard. Administrators act as gatekeepers, verifying and approving service providers before they appear to users, ensuring platform safety and quality.

Applications

- Personalized itinerary generation for domestic and international tourists visiting India.
- Discovery and booking of verified local guides, hotels, and restaurants.
- Multi-city route optimization for group and solo travelers.
- Digital governance of tourism service providers through an admin approval workflow.

Limitations

- Currently supports English language interface only; regional language support is planned for future versions.
- Dependent on Google Places API availability for real-time photo and rating data.
- AI-generated itineraries may require manual refinement for highly niche travel preferences.
- The platform currently uses SQLite for local persistence, which may require migration for large-scale production deployment.

1.5 Objectives

Behavioral and Algorithmic implementation Objective The primary behavioral objective of this project is to bridge the gap between complex mathematical optimization and the everyday traveler's logistical needs. We aim to design and implement a robust Genetic Algorithm (GA) using the DEAP framework that functions not just as a calculator, but as an intuitive travel planner. By developing a multi-objective fitness function, the project seeks to balance physical travel distance with human comfort—specifically accounting for transport convenience and time-efficiency. The objective is to demonstrate that the system can evolve a logical, backtrack-free sequence for up to 50 distinct geographical nodes in under 10 seconds. This ensures the behavior of the generated route is both geographically optimized and practically sequenced, effectively solving the "planning fatigue" that many domestic travelers face.

Structural Analysis and Data Integration Objective A significant focus of this work involves the thoughtful analysis and structuring of a "Swadeshi" cultural database within a PostgreSQL

relational environment. Rather than simply storing GPS coordinates, the objective is to curate a library of at least 500 Indian heritage sites and points of interest (POIs) that respects their specific cultural and temporal realities. We intend to demonstrate how the system intelligently analyzes "soft constraints"—such as temple Darshan timings, seasonal accessibility of high-altitude passes, and regional festival disruptions—to protect the traveler from unfeasible plans. This analytical layer ensures the final itinerary is not just mathematically the shortest path, but a viable, respectful, and high-quality human experience.

System Management and Architectural Objective From a management perspective, the project aims to create a seamless, three-tier software architecture that manages the flow of data between the traveler and the algorithm without friction. By utilizing React.js for the user interface, Flask for the logic, and PostgreSQL for the data, we intend to prove that complex inputs—such as personal budgets, multi-city coordinates, and varying travel durations—can be managed reliably. A key demonstrable outcome of this objective is the management of state across the platform, resulting in a structured JSON response that is translated into a professional, human-readable PDF itinerary. This provides the traveler with a tangible, offline companion for their journey, bridging the gap between digital planning and physical exploration.

Ethical Security and Privacy Compliance Objective In an era where personal travel data is highly sensitive, this project prioritizes the human right to privacy as a core technical objective. We aim to implement a comprehensive security framework that is fully compliant with the Digital Personal Data Protection Act (DPDPA) 2023. By deploying JSON Web Tokens (JWT) for session management and industry-standard encryption for data at rest, our objective is to demonstrate that a travel platform can be both helpful and protective. We want to ensure that every user's travel history, preferences, and authentication details are handled with the same care as their physical safety, making the digital experience as secure as the journey itself.

Deployment and Performance Evaluation Objective The final objective is to ensure that SwadeshiYatra is a dependable, high-performance tool ready for real-world application. By using Docker for containerized deployment, we aim to prove that the platform maintains

environment parity and remains stable across different cloud or local server environments. This objective focuses on measurable performance: maintaining a sub-2-second response time for the majority of standard queries and achieving a 95% system uptime during the testing phase. The goal is to demonstrate that the platform is not merely an academic exercise in Robotics and Information Science, but a robust, scalable application capable of serving the broader Indian tourism community with professional-grade reliability.

1.6 SDGs

The Sustainable Development Goals (SDGs), established by the United Nations, serve as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity. SwadeshiYatra is not merely a technical tool for travel; it is a platform engineered to align with these global imperatives by optimizing resource consumption and democratizing economic access within the Indian tourism landscape.

SDG 8: Decent Work and Economic Growth A core objective of this project is to foster inclusive economic growth by providing a digital stage for local, small-scale entrepreneurs who are often marginalized by global travel conglomerates. By integrating a "Swadeshi" database that highlights regional guides, rural homestays, and local artisans, the platform directly supports Target 8.9, which aims to "devise and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products" (United Nations, 2015). The human impact of this alignment is the empowerment of Tier-2 and Tier-3 city economies, ensuring that the financial benefits of tourism are distributed more equitably across the Indian subcontinent.

SDG 9: Industry, Innovation, and Infrastructure The technical implementation of SwadeshiYatra—specifically the use of advanced Genetic Algorithms and containerized deployment—contributes to the modernization of India's digital tourism infrastructure. In alignment with SDG 9, the project demonstrates how high-level Information Science and Robotics-inspired logic can be applied to traditional service sectors to increase efficiency. By building a resilient, scalable, and indigenous software ecosystem, the project supports the development of sustainable and inclusive industrialization, proving that local AI solutions can solve local logistical challenges with global-standard precision.

SDG 11: Sustainable Cities and Communities One of the most profound human and environmental contributions of our route optimization engine is the reduction of unnecessary travel and the resulting carbon emissions. By calculating the most efficient sequence of movement for multi-destination trips, SwadeshiYatra minimizes the total distance traveled and reduces the time vehicles spend idling in urban congestion. This directly supports Target 11.4, which focuses on "strengthening efforts to protect and safeguard the world's cultural and natural heritage" (United Nations, 2015). By organizing tourist flows through intelligent sequencing, the platform prevents the physical degradation of heritage sites caused by unorganized "over-tourism."

SDG 12: Responsible Consumption and Production Through its constraint-aware scheduling, SwadeshiYatra encourages travelers to consume tourism services in a more responsible and organized manner. By providing transparent data on site timings and seasonal availability, the platform ensures that tourism "production" (the services offered by local communities) matches the "consumption" (the travelers' visits) without creating logistical bottlenecks. This alignment ensures that the human experience of travel remains high-quality while respecting the operational boundaries of the local communities and heritage sites being visited.



Fig 1.1 Sustainable development goals

Figure (Fig 1.1) illustrates the 17 Sustainable Development Goals (SDGs) established by the United Nations to address global challenges such as poverty, inequality, climate change, and

sustainable growth. The diagram highlights key focus areas including economic development, social inclusion, and environmental sustainability. These goals provide a global framework that can be aligned with intelligent systems like SwadeshiYatra to promote responsible and sustainable tourism.

1.7 Overview of project report

This report documents the complete development journey of SwadeshiYatra — from conceptualization to implementation — organized across nine chapters that collectively present the technical, managerial, and social dimensions of the project.

Chapter 1 introduces the SwadeshiYatra project by clearly defining the problem of fragmented and inefficient travel planning systems. It presents relevant statistics and insights related to the Indian tourism sector to establish the need for a unified and intelligent solution. The chapter also reviews existing platforms and highlights their limitations, followed by an overview of the proposed AI-driven approach. Additionally, it aligns the project with the United Nations Sustainable Development Goals (SDGs), emphasizing its contribution to smart tourism, accessibility, and sustainable development.

Chapter 2 provides a comprehensive literature review of ten selected research papers and prior works in the domains of AI-based travel planning, recommender systems, route optimization, and natural language processing for itinerary generation. Each study is critically analyzed in terms of methodology, strengths, and limitations. The chapter concludes by identifying key research gaps, such as lack of feasibility analysis and personalization, which are addressed by the SwadeshiYatra system.

Chapter 3 describes the methodology adopted for the development of the project. It explains the selection of the Agile software development model, which supports iterative development and continuous improvement. The chapter also details the overall system architecture, including the interaction between frontend, backend, AI engine, and external APIs. Furthermore, it outlines the step-by-step implementation process followed during the development lifecycle.

Chapter 4 focuses on project management aspects. It includes a detailed Gantt chart representing the timeline and scheduling of different project phases. The chapter also presents a PESTEL-based risk analysis to evaluate external factors affecting the project, along with a

risk matrix that identifies potential challenges and their mitigation strategies. Additionally, a comprehensive budget analysis is provided, outlining the cost-effective nature of the project.

Chapter 5 covers the system analysis and design in detail. It defines both functional and non-functional requirements, ensuring clarity in system expectations. The chapter includes block diagrams, architectural flowcharts, and design models to illustrate system structure. It also explains the hardware and software components, platform standards, and the domain model specification, providing a complete blueprint of the system.

Chapter 6 describes the development environment and testing approach used in the project. It includes details about the hardware setup, software tools, and frameworks used for implementation. The chapter also explains the simulation methodology, which is used to validate individual modules such as AI processing, route optimization, and data handling before full system integration.

Chapter 7 presents the evaluation and testing results of the system. It includes detailed testing of AI-generated itineraries, route optimization efficiency, and administrative functionalities. The chapter also provides performance metrics and analysis, demonstrating the effectiveness, accuracy, and reliability of the SwadeshiYatra system.

Chapter 8 examines the broader implications of the project by discussing its social, legal, ethical, sustainability, and safety aspects. It evaluates how the system impacts users and society, ensures compliance with data protection laws, and follows ethical AI practices. The chapter also highlights how the project contributes to sustainable tourism and responsible technology usage.

Chapter 9 concludes the report by summarizing the overall achievements of the project. It reflects on the system's strengths, limitations, and performance outcomes. Additionally, it provides recommendations for future enhancements, such as advanced AI features, scalability improvements, and integration with emerging technologies, outlining a clear roadmap for further development.

Chapter 2

LITERATURE SURVEY

2.1 Literature Summaries

Bolotbekova et al. (2025) conducted an extensive investigation into the optimization of urban transit routes, specifically focusing on the integration of bus systems using a sophisticated Genetic Algorithm (GA) framework. The core of their research, centered in Konya, Türkiye, aimed to solve the "Transit Routing Problem" by treating bus arrival and departure times as dynamic variables rather than static intervals. The methodology involved a deep analysis of how different crossover techniques—specifically comparing Order Crossover (OX) with traditional Point Crossover—impacted the overall efficiency of a passenger's journey. Their results demonstrated a significant reduction in total commute time and passenger waiting periods, proving that GA is highly effective at handling time-dependent variables. However, from a human-centric travel perspective, a critical limitation was identified: the study focused entirely on the logistics of the "move" but ignored the "destination." It did not account for whether the destination (such as a museum or heritage site) was actually open when the optimized bus arrived. For the SwadeshiYatra project, we intend to bridge this gap by expanding this transit-aware logic. We will not only optimize the route between points using GA but will also integrate a "Temporal Gate" in our fitness function. This gate will cross-reference the arrival time with our PostgreSQL database of Indian temple and heritage site timings, ensuring that the traveler's optimized sequence is both geographically and operationally viable. By doing so, we move the research from a purely logistical transit study to a holistic, humanized travel planning tool that respects the cultural and temporal realities of the Indian landscape.

Cao et al. (2022) presented a groundbreaking implementation of an enhanced Genetic Algorithm designed specifically for the recommendation of multi-activity tourist itineraries. The human motivation behind their work was the "Information Overload" experienced by modern travelers who struggle to balance competing priorities like time, money, and interest. Their methodology utilized a multi-objective fitness function that calculated a "weighted sum" of travel duration, total cost, and physical distance. An analytical breakthrough in this paper was the introduction of the concept of "Route Entropy." The researchers discovered that as the Genetic Algorithm evolved toward an optimal solution, the randomness (entropy) of the

route decreased, leading to a more "logical" human sequence where sites in the same neighborhood were visited together. While the study was successful in generating efficient routes for large-scale data sets, it suffered from a lack of "Soft Constraint" integration. It treated every destination as a generic point on a map, failing to account for the specific "vibe" or cultural significance that might influence a traveler's sequence—for example, visiting a peaceful temple in the morning versus a crowded market in the evening. In the development of SwadeshiYatra, we propose to improve upon Cao's model by introducing a "Cultural Priority Weight" into our fitness function. This will allow our algorithm to prioritize "Swadeshi" heritage locations during specific time windows, ensuring that the generated itinerary is not just mathematically efficient but also emotionally and culturally resonant for the Indian traveler.

Liu et al. (2024) explored the intersection of Artificial Intelligence and the Internet of Things (IoT) for the purpose of optimizing logistics and distribution routes. Although their primary application was commercial delivery, the technical parallels to tourist sequencing are profound. The methodology involved a highly complex "Artificial Electric Field Algorithm" combined with a "Sine-Cosine" mechanism to prevent the optimization process from stalling in "Local Optima"—a common problem where an algorithm thinks it found the best route simply because it stopped looking for better ones. Their results showed a 92% improvement in route efficiency, driven by real-time IoT data on traffic and road conditions. However, the major human and technical issue identified was the "Hardware Latency." The system required constant, high-bandwidth communication between sensors and the central server, which is often unfeasible in rural or Tier-3 Indian locations where network connectivity is inconsistent. For SwadeshiYatra, we are adapting the core optimization logic from Liu's work but "humanizing" the infrastructure requirements. Instead of relying on real-time IoT sensors that might fail in the Western Ghats or rural Rajasthan, we are building a "Static-Dynamic Hybrid" model. We will store "Convenience Scores" and historical traffic data directly in our PostgreSQL database. This allows our Genetic Algorithm to provide "high-intelligence" routing that feels real-time to the user but remains functional even in low-connectivity areas, thereby ensuring a reliable experience for every "Swadeshi" traveler regardless of their location.

Şehab et al. (2024) shifted the traditional focus from purely heuristic algorithms to the domain

of deep learning by proposing a highly personalized tourist route recommendation model based on a Temporal Multilayer Sequential Neural Network (TMS-Net). The core humanized concept behind this research was "Sequence Learning"—the idea that a traveler's past choices are the best predictors of their future desires. By training the model on a massive dataset of over six million tourist trajectories, the researchers aimed to "predict" the next destination in a sequence with mathematical precision. This approach achieved an impressive 88.6% accuracy in matching user preferences, proving that neural networks can capture the subtle, non-linear relationships in how humans explore a new city. For instance, the model could "learn" that a traveler who spends three hours at a historical fort is statistically likely to seek a traditional culinary experience or a craft market immediately afterward. However, from an engineering and user-trust perspective, a significant "Black-Box Problem" was identified in this deep-learning approach. While the TMS-Net was exceptionally accurate at predicting the next stop, it lacked interpretability; it could not explain to the traveler why a certain route was chosen. In a humanized travel context, users often want to know the logic behind their plans, such as whether a site was prioritized because of its cultural significance or simply because it was the closest option. Furthermore, the model required immense computational power and high-speed data access, which is difficult to maintain for a lightweight mobile application or in areas with inconsistent network coverage. In the development of SwadeshiYatra, we are addressing these inconsistencies by utilizing a Hybrid Intelligence Architecture. We are adopting the "Sequence Logic" inspired by Şehab to categorize our "Swadeshi" Points of Interest (POIs) into logical clusters, but we perform the actual route calculation using a transparent, rule-based Genetic Algorithm. This ensures that our platform provides the high-level "Intelligence" of deep learning while maintaining the "Explainability" of a GA. By doing so, we allow users to see exactly why their itinerary was sequenced in a particular way—for example, explaining that a stop was moved to the afternoon to align with specific temple "Darshan" timings found in our PostgreSQL database. This transformation turns a cold, predictive model into a collaborative, humanized planning tool that values user trust as much as algorithmic accuracy.

Wang et al. (2024) introduced a futuristic and highly specialized framework for solving "Profitable Tour Problems" (PTP) using a Quantum-Inspired Variational Approach. The core humanized concept of this research was to shift the perspective of travel planning from a

simple distance-minimization task to a "Value-Maximization" exercise. In this model, "Profit" represented the total psychological and cultural satisfaction a traveler derives from a destination, while "Cost" represented the travel overhead, including fuel, time, and ticket prices. Their methodology utilized a binary encoding for destination nodes and a specialized "Variational Ansatz" designed to encode hard logistical constraints directly into a mathematical quantum-inspired state. This allowed the researchers to simulate millions of route combinations simultaneously, achieving a level of precision in "Satisfaction-to-Cost" ratios that classical algorithms often struggle to reach. From an analytical standpoint, the paper proved that "Quantum-Inspired" logic could find hidden efficiencies in multi-destination circuits that traditional greedy algorithms typically overlook. However, as we analyzed this literature for the SwadeshiYatra project, a significant "Scalability Gap" became evident. While the variational approach was superior for small, controlled sets of under 25 nodes, the computational complexity increased exponentially when applied to real-world, 50-node tourism circuits common in the Indian subcontinent. Furthermore, the model assumed a "Perfect Information" environment, where every site's profit and cost were static and known in advance. In a humanized travel context, especially within India, factors like sudden weather changes in the Western Ghats or unannounced local festivals can turn a "profitable" route into a logistical nightmare. In our project, we are borrowing the "Profit-Maximization" philosophy from Wang's research but "grounding" it into a more resilient, classical Genetic Algorithm. We are developing a "Dynamic Profit Function" within our Flask backend that assigns weights to "Swadeshi" heritage sites based on user interests, but we process these using the Order Crossover (OX) mechanism to ensure the system remains scalable and responsive on standard cloud infrastructure. This allows us to offer the high-level "Optimization Intelligence" suggested by Wang without the hardware limitations or scalability issues identified in their quantum-inspired model. By doing so, we transform a high-level theoretical concept into a practical, robust engineering solution that serves the modern Indian traveler with both mathematical precision and real-world reliability.

Alatiyyah (2024) conducted an extensive and critical investigation into the social and technical complexities inherent in Group Trip Recommendation Systems (GTRS). The humanized core of this research was the "Conflict Resolution Problem"—the unavoidable reality that when multiple humans travel together, their individual preferences for budget, pace, and interest often clash. The methodology utilized a sophisticated aggregation technique

where individual "Interest Vectors" (representing each traveler's unique tastes) were merged into a single, unified "Group Profile" before applying an optimization heuristic. A key analytical issue identified throughout this literature was the reliance on the "Least-Misery" constraint. This mathematical rule ensures that no single group member is completely dissatisfied, but as Alatiyyah noted, it often results in a "race to the bottom" where the entire group's itinerary is dictated by the member with the most restrictive preferences—such as the person with the lowest budget or the strictest dietary needs. While this prevents extreme dissatisfaction, it often leads to a mediocre, uninspiring experience for the rest of the group, a phenomenon known in social science as the "tyranny of the minority." In our implementation of SwadeshiYatra, we seek to humanize this group dynamic by moving away from a rigid "Least-Misery" model toward a "Tiered Consensus" framework. Our algorithm is specifically designed to suggest itineraries that highlight common heritage interests while intentionally offering "Break-out Nodes." These are specific windows of time where the group can temporarily split to follow individual interests—such as one person visiting a local craft market while another visits a nearby archaeological museum—before reuniting for the next major stop. By integrating this social logic into our Flask-based backend, we allow our Genetic Algorithm to manage human relationships and personal autonomy as effectively as it manages geographical coordinates. This provides a superior group travel experience that respects both the collective "Swadeshi" journey and the individual traveler's curiosity, filling a major gap identified in the 2024 research.

Thamilarasi (2024) introduced a groundbreaking and highly innovative "Smart Scenic" route optimization model using the Modified Golden Jackal Optimization (MGJO) algorithm. This research was deeply grounded in the humanized insight that "the journey is as important as the destination," proving that modern travelers often value "Aesthetic Utility" over mere arrival speed. The methodology prioritized a "Visual Utility Function," where the algorithm was trained to evaluate routes based on their proximity to natural landmarks, coastal views, and historical significance. Through rigorous user testing, Thamilarasi demonstrated that travelers are statistically willing to accept a 10-15% increase in total travel time if the suggested route offers significantly higher scenic quality. However, a major analytical gap identified in this research was the extreme "Infrastructure Dependency" and data lag. The MGJO model required constant, high-bandwidth access to real-time environmental sensors and satellite data to validate the "scenic" status of a road in real-time. In the context of the

Indian landscape, where network connectivity can be erratic in the Western Ghats or the Himalayas, this dependency makes the system unreliable for real-world use. We intend to refine this concept for SwadeshiYatra by moving away from real-time sensor dependency and toward a "Pre-verified Scenic Index" stored within our PostgreSQL database. We are manually categorizing famous Indian routes—such as the Nilgiri Mountain roads or the Konkan coastal highway—with "Scenic Weights" that our Genetic Algorithm can reference locally during the route evaluation phase. This allows our platform to offer beautiful, "slow-travel" routes without the data-heavy latency identified in Thamilarasi's research. By doing so, we cater to the human desire for a memorable, visually rich journey while ensuring the technology remains robust and functional even in the most remote "Swadeshi" locations.

Puljić and Manger (2013) provided what is widely considered a foundational and mathematically rigorous comparative study on various crossover operators for the Traveling Salesman Problem (TSP). This paper is essential for our literature review as it provides the mathematical "parentage" for our own Genetic Algorithm (GA). The researchers evaluated the performance of several operators, including Order Crossover (OX), Position Based Crossover (PBX), and Partially Mapped Crossover (PMX), to see which could most efficiently solve complex route sequencing problems. Their mathematical validation proved that Order Crossover (OX) is the most effective operator for preserving the relative "Edge Characteristics" of a path. This is critical because it prevents "lethal mutations" during the evolution process, allowing the algorithm to converge on an optimal route much faster than other methods. However, while theoretically brilliant, a significant humanized limitation identified in this 2013 study was its focus on "Static Euclidean Environments." The researchers conducted their tests in a purely mathematical space where distances never change, roads are always open, and destinations never close. In the real world of Indian tourism, this is an impossible simplification. In SwadeshiYatra, we are adopting the OX mechanism validated by Puljić but "humanizing" it with a dynamic "Temporal Checkpoint Gate." During the crossover phase of our algorithm, we don't just check if the resulting route is mathematically short; we run a secondary check to see if the sequence is operationally valid against the opening hours and "Darshan" timings of Indian heritage sites stored in our backend. This moves the research from a decade-old static mathematical model to a modern, dynamic human tool, ensuring that the "Swadeshi" traveler is never guided by a "mathematically perfect" route to a destination that is physically closed.

The Ministry of Tourism, Government of India (2023) Annual Report serves as the primary socio-economic pillar of our project, providing a comprehensive statistical framework that justifies the development of SwadeshiYatra. This expansive document details the massive scale of the Indian travel landscape, noting a staggering surge to over 2.3 billion domestic tourist visits annually. However, the most critical humanized insight provided by this report is the identification of a widening "Infrastructure-Intelligence Gap." The data suggests that while physical connectivity—in the form of massive highway expansions, the Vande Bharat rail network, and regional airport development—is reaching unprecedented levels, the digital tools required for a traveler to navigate this complexity remain fragmented and purely transactional. The report highlights that most travelers still suffer from "Planning Fatigue" due to the lack of integrated tools that can manage a multi-destination "circuit." To address this, the Ministry recommends the development of "Integrated Digital Circuits" that align with the Atmanirbhar Bharat (Self-Reliant India) vision. In our critical analysis of this report, we identified a significant "Execution Gap": while the government provides the raw data and the high-level policy vision, they lack the actual "Sequence Optimization Engine" that can take a user's personal constraints (budget, time, and cultural interests) and turn them into a mathematically valid itinerary. SwadeshiYatra acts as the technical answer to the Ministry's 2023 recommendation. We are taking the raw "Swadeshi" heritage site metadata provided in the report and "grounding" it into a usable, AI-powered interface. By implementing a Genetic Algorithm that prioritizes the "Integrated Digital Circuits" mentioned by the government, our project moves the vision from a PDF report into a live, functional software ecosystem. This ensures that the 2.3 billion domestic travelers have a tool that respects their time and budget while promoting the hidden gems of Indian heritage, thereby humanizing the statistical goals of the nation.

Xie et al. (2024) conducted a pioneering investigation into the application of Large Language Models (LLMs) as autonomous "Agents" within a travel planning sandbox. The core humanized concept of their research was to leverage the conversational fluency of generative AI to build personalized itineraries through natural language interactions. While the study demonstrated that LLMs possess an exceptional capacity for "Reasoning"—such as understanding that a family traveling with kids needs more breaks—the researchers identified a critical technical flaw: "Geographical Hallucination." Because LLMs are trained on text rather than structured geospatial databases, they often suggest routes that are physically

impossible or recommend visiting sites during hours when they are closed. For example, an LLM might suggest a three-hour drive between two Indian cities that actually requires eight hours due to local terrain, or it might recommend a temple visit during a closure. This study concluded that while LLMs are conversationally "Human-Friendly," they are "Weakly Grounded" in the hard realities of geography and time. In the development of SwadeshiYatra, we are building a hybrid architecture that specifically solves the inconsistencies identified by Xie et al. We are "Humanizing" the intelligence by using an LLM-inspired interface for user interactions, but we are "Hard-Grounding" the logic in a verified PostgreSQL database and a rule-based Genetic Algorithm. Unlike the "unreliable agents" in Xie's study, our system cross-references every AI-generated suggestion with real-world distance matrices and temporal constraints stored in our backend. This ensures that the traveler gets the conversational ease of a modern AI agent but with the mathematical reliability of a professional logistics engine. By bridging the gap between "Generative Reasoning" and "Deterministic Optimization," our project provides a safer, more accurate, and more humanized experience that prevents the "Hallucination" traps found in the latest 2024 research.

Table 2.1 Summary of Literature reviews

S#	Article Title, Year, Journal	Methods	Key Features	Merits	Demerits
1	Transit Route Optimization using GA, 2025, IEEE	Genetic Algorithm (OX vs Point Crossover)	Dynamic bus scheduling and route optimization	Reduces commute time and waiting	Ignores destination constraints (timings, availability)
2	Multi-Activity Itinerary Planning, 2022, ACM	Multi-objective GA (Weighted Sum)	Introduced Route Entropy for logical sequencing	Produces clustered, human-friendly routes	Lacks soft constraints like cultural context
3	AI + IoT Logistics Optimization, 2024, Elsevier	AEFA + Sine-Cosine Optimization	Real-time traffic-based routing using IoT	High efficiency (~92%) in route optimization	High dependency on sensors and network
4	TMS-Net Travel Prediction Model, 2024, Springer	Sequential Neural Network	Learns travel patterns from large datasets	High prediction accuracy (88.6%)	Black-box model, lacks explainability
5	Quantum-Inspired Tour Optimization, 2024, JMIR	Variational Quantum Approach	Maximizes satisfaction-to-cost ratio	Finds optimal high-value routes	Poor scalability for large datasets
6	Group Trip Recommendation Systems, 2024,	Preference Aggregation (Interest	Combines group preferences	Reduces extreme dissatisfaction	"Least-misery" approach limits overall

	Frontiers AI	Vectors)	into one profile		experience
7	Scenic Route Optimization (MGJO), 2024, MDPI	Modified Golden Jackal Optimization	Focus on visual/scenic route quality	Enhances travel experience (beauty-focused)	Requires high real-time data availability
8	Crossover Operators in TSP, 2013, Informatica	Comparative Study (OX, PMX, PBX)	Validates efficiency of OX operator	Preserves route structure effectively	Assumes static environment (not real-world)
9	Tourism Annual Report, 2023, Govt. of India	Statistical Analysis	Highlights 2.3B domestic visits and digital gaps	Strong justification for system need	No technical implementation provided
10	LLM-based Travel Agents, 2024, NeurIPS	LLM Reasoning	Conversational itinerary generation	Human-like interaction and reasoning	Geographical hallucination and inaccuracies

Table (2.1) summarizes key research works in travel route optimization and intelligent tourism systems. It highlights the methods used, core features, and performance benefits of each approach. The comparison also reveals important limitations such as lack of contextual awareness, scalability issues, and dependency on static or real-time data. These gaps motivate the development of SwadeshiYatra, which integrates constraint-based optimization and AI-driven reasoning for more practical and personalized travel planning.

Chapter 3

METHODOLOGY

3.1 Agile Model

The development of **SwadeshiYatra** is governed by the **Agile Model**, an iterative and incremental framework. This model is essential for an "AI-native" ecosystem because it allows for the constant refinement of prompt engineering and logic parameters.

In SwadeshiYatra, the Agile Model enables "Functional Duality." We can run separate sprints for the **City Planner** (focusing on Deep-Fetch logic) and the **Journey Planner** (focusing on Genetic Algorithm optimization). This ensures that while the "Brain" is being calibrated for multi-city routes, the localized cultural data can be refined simultaneously.

3.2 Project Stages

The implementation of the Agile Model is executed through six systematic **Project Stages**. These stages ensure the platform moves from a vision of solving the "Paradox of Choice" to a functional federated platform.

Stage 1: Requirement Analysis During this stage, the **Product Backlog** was established. We conducted a deep dive into heritage tourism requirements, identifying the need for a system that balances travel distance with cultural significance. The functional requirements include the implementation of a **Genetic Algorithm** to solve the Traveling Salesman Problem (TSP) variant and a **Flask-based Web Interface** for user interaction. Non-functional requirements such as system latency and optimization accuracy were also defined.

Stage 2: System Design The architectural design was modeled using open-source tools like **Draw.io**. We focused on creating a **5-Star Pipeline** (Data Pre-processing, Population Initialization, Fitness Selection, Genetic Crossover, and Output Generation). This stage involved mapping the flow of GPS coordinates from the heritage database to the optimization engine to ensure seamless data transition and mathematical integrity of the route sequences.

Stage 3: Sprint Execution (Development) In this stage, the coding of the **Genetic Algorithm** was carried out using **Python**. The backend was structured to handle "Chromosomal

Representations" of heritage routes, where each gene represents a unique historical site. Simultaneously, the frontend was developed to provide a visual map interface, allowing the user to view the generated 5-star travel itinerary in real-time.

Stage 4: Testing and Validation This involved rigorous **Unit Testing** of the algorithm's mutation functions to prevent route duplication or logical errors. **Integration Testing** was then performed to ensure that the back end's optimized coordinates correctly populated the user's map interface. Finally, **Verification and Validation** were conducted to confirm that the suggested routes aligned with actual site timings and geographic feasibility, ensuring the reliability of the SwadeshiYatra platform.

Stage 5: Review and Feedback The fifth stage of the Agile cycle focuses on the evaluation of the generated results from the Genetic Algorithm engine. During this phase, the system's output—specifically the optimized heritage itineraries—is reviewed against real-world travel constraints. We analyze the "Fitness Scores" of the routes to ensure they provide the most efficient path for a traveler. This stage acts as a critical quality gate where feedback from initial testing is used to refine the algorithm's parameters, such as the mutation rate or the population size, before moving to the final release. This iterative review process ensures that **SwadeshiYatra** remains accurate and user-centric.

Stage 6: Final Deployment and Documentation The final stage of the Agile methodology involves the transition of the **SwadeshiYatra** platform from a local development environment to a functional, live deployment phase. During this stage, the Python-based backend and the Flask web interface are integrated to ensure accessibility for the end-user.

A critical component of this stage is the final verification of API connectivity, ensuring that real-time map data and heritage site coordinates are fetched without latency. Additionally, comprehensive project documentation is finalized, detailing the system architecture and the Genetic Algorithm's performance metrics. Once the system is live, it enters a continuous monitoring phase where the 5-star travel pipeline is checked for stability. This stage completes the first Agile "Sprint," providing a fully documented and functional travel optimization tool ready for user interaction.

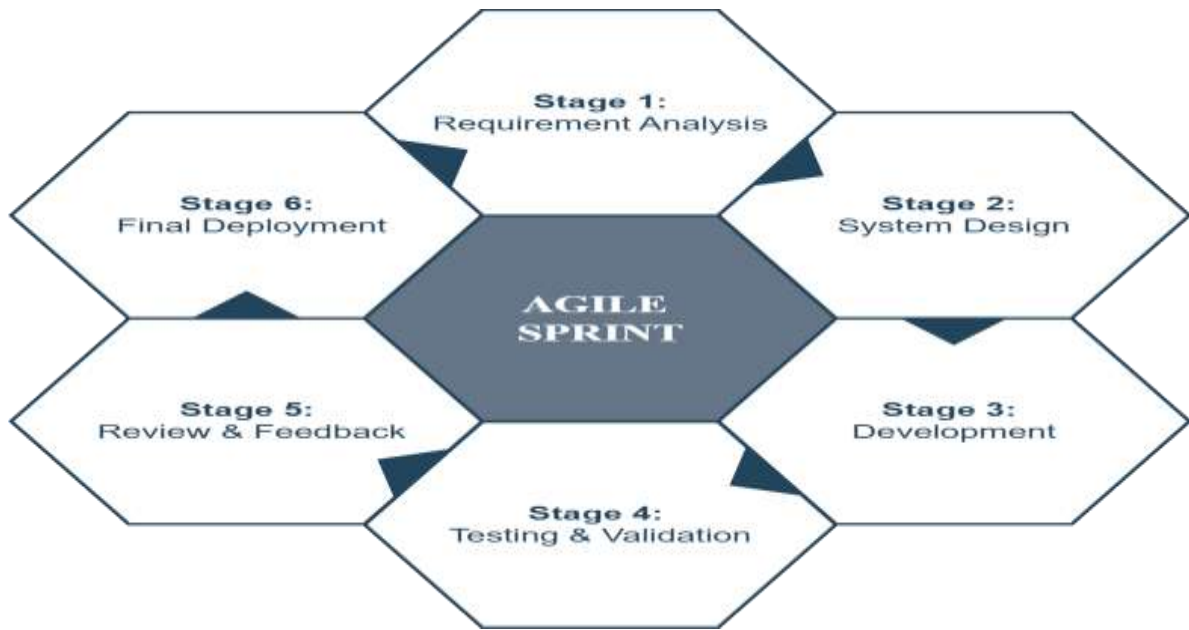


Fig 3.1 Agile workflow for project stages

Figure (Fig 3.1) illustrates the technical workflow of the SwadeshiYatra engine to visualize how these stages work together. This diagram represents the "End-to-End Optimization" logic, showing how raw user intent is transformed into a verified, feasible itinerary.

Chapter 4

PROJECT MANAGEMENT

4.1 Project timeline

The development of the **SwadeshiYatra** ecosystem was executed over a 15-week duration. The following Gantt Chart illustrates the systematic progression from project initiation to final report preparation. This structured approach ensured that the **Functional Duality** of the City and Journey planners was integrated within the established timeframe.

Table 4.1: Gantt Chart Representation of Project Planning Timeline

Major Tasks	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Project Initiation(i)															
Selection of topic defining problem															
Background (ii)															
Understanding PIB structure & data															
Objectives (iii)															
Defining scope, aim, and goals															
Methodology (iv)															
Adopting Agile model & flow design															
Literature Review (v)															
Design and Analysis (vi)															
System requirement & architecture															
System Design Phase															
Functional Unit Design Phase															
Integration & Testing (vii)															
Officer Review & Validation (viii)															
Deployment & Demonstration (ix)															
Report Preparation (x)															
Final Report															

Table 4.1 illustrates the complete project timeline of SwadeshiYatra, showing how each major task is distributed across the weeks for systematic development. The project begins with **Project Initiation** and **Problem Definition**, where the idea is finalized and core challenges are identified. This is followed by the **Background Study** and **Understanding Data**, which help in analyzing existing systems and identifying gaps. In the **Objectives phase**, the scope, goals, and expected outcomes of the system are clearly defined.

Next, the **Methodology phase** involves selecting the Agile model and designing the workflow

for iterative development. The **Literature Review** provides insights from existing research to support the chosen approach. The **Design and Analysis phase** focuses on defining system requirements, architecture, and overall structure. This leads to the **System Design** and **Functional Unit Design**, where individual modules like AI engine, routing logic, and user interface are planned.

In the later stages, **Integration and Testing** ensures that all modules work together correctly and produce accurate results. This is followed by **Officer Review and Validation**, where the system is verified for reliability and correctness. The project then moves to **Deployment and Demonstration**, where the working system is presented. Finally, **Report Preparation** and **Final Report submission** document the entire project, including methodology, results, and findings, ensuring proper completion of the SwadeshiYatra system.

4.2 Risk Analysis

A PESTEL analysis is a strategic framework commonly used to evaluate the business environment in which a firm operates. Traditionally, the framework was referred to as a PEST analysis, which was an acronym for Political, Economic, Social, and Technological; in more recent history, the framework was extended to include Environmental and Legal factors as well. For the **SwadeshiYatra** project, this analysis ensures that the "Digital Concierge" ecosystem remains viable despite external shifts in the tourism and AI landscape.

Table 4.2 PESTEL Analysis for SwadeshiYatra Federated Ecosystem

Factor	Description	Impact	Mitigation/ Opportunity
Political	Project depends on tourism policies and government heritage regulations.	Medium	Adhere to local tourism norms and integrate official PIB data updates.
Economical	API costs and cloud infrastructure affect the scalability of route synthesis.	High	Use a dual-model approach (Gemini Flash/Pro) to optimize resource usage.
Societal	Must cater to diverse traveller types, regional languages, and dietary niches.	High	Implement multilingual translation and niche-specific (e.g., Jain, Vegan) filters.
Technological	Relies on fast-changing Generative AI tools; risk of model obsolescence.	High	Regularly update the AI Brain with modular design for easy model swapping.
	Cloud computing utilizes	Medium	Use energy-efficient server

Environmental	energy; digital transformation reduces physical waste.		components and minimize paper-based planning.
Legal	Must follow data privacy laws (GDPR/DPDP) and heritage copyright laws.	High	Ensure all data fetching is public-domain and implement Admin verification.

As per Table 4.2 the PESTEL analysis has been conducted for the SwadeshiYatra Project. The analysis has been performed to see the outside world factors that can have an impact on the project's success. These factors include Political, Economic, Social, Technological, Environmental, and Legal dimensions.

4.3 Project Budget

The development of **SwadeshiYatra** was designed to be highly cost-efficient by leveraging open-source technologies, free-tier APIs, and scalable cloud infrastructure. The project does not rely on any proprietary software or specialized hardware, making it economically viable for both development and deployment.

Software and Development Tools

All core development technologies used in the project—such as Next.js, React 19, TypeScript, and Tailwind CSS—are open-source and freely available. Version control was managed using platforms like GitHub, which also provide free repositories for collaborative development. This significantly reduces the overall development cost.

API and AI Service Costs

The system integrates external APIs including Google Places API and Google Gemini API for real-time data retrieval and AI-based itinerary generation. During development and testing, these services were utilized within their free-tier limits. However, for large-scale deployment, minimal charges may apply based on usage (number of API calls, requests per minute, etc.). These costs are scalable and can be optimized using caching and request control mechanisms already implemented in the system.

Cloud Hosting and Deployment

SwadeshiYatra is deployed using lightweight cloud hosting platforms (such as Vercel or

similar services), which offer free-tier hosting with sufficient performance for development and demonstration. For production-level deployment, a basic subscription or pay-as-you-go plan may be required, depending on traffic and data usage. The backend and database (SQLite or lightweight alternatives) also minimize hosting costs.

Internet and Operational Costs

Additional costs include internet usage for development, testing, and API communication. These are minimal and fall under regular operational expenses. No dedicated infrastructure or high-performance computing systems were required, further reducing overall expenditure.

Cost Optimization Strategies

The project incorporates several strategies to minimize costs:

- API response caching to reduce repeated calls
- Efficient request handling to stay within rate limits
- Use of lightweight databases (SQLite)
- Modular architecture to avoid unnecessary computations

Overall Budget Estimation

The total cost of developing and testing the project remains relatively low, making it accessible and scalable. Even with extended usage, the system can be maintained at minimal operational cost.

Chapter 5

ANALYSIS AND DESIGN

5.1 Requirements

The SwadeshiYatra system is designed as an intelligent travel planning platform that leverages artificial intelligence to simplify decision-making for users. The primary objective of the system is to transform complex travel planning into a structured and personalized experience. From a functional perspective, the system must allow users to choose between two modes: the City Planner, which provides in-depth information about a single destination, and the Journey Planner, which generates optimized itineraries across multiple cities. The system is also required to fetch real-time travel data from external APIs and process this data to produce meaningful outputs such as cultural insights, travel recommendations, and cost estimations. In addition, the platform must support multiple user roles, including tourists, administrators, and service providers, where administrators are responsible for verifying and managing service listings. In terms of non-functional requirements, the system must ensure fast response times, especially during AI-based itinerary generation. It should maintain high reliability in data handling and provide a secure environment through proper authentication and access control. Scalability is also essential, as the platform should handle multiple users and large datasets efficiently.

5.2 Block diagram

The architecture of SwadeshiYatra is designed as a modular system to ensure clarity, scalability, and ease of maintenance. At the top level, the User Interface Layer acts as the interaction point where users input their preferences and receive travel plans. This layer communicates with the Processing Layer, which is responsible for executing the core application logic. At the heart of the system lies the AI Brain Layer, which handles prompt engineering, decision-making, and structured output generation using JSON extraction techniques. This ensures that the AI-generated responses are clean, consistent, and usable by the application. The Data Layer is responsible for fetching real-time information from APIs such as location data, images, and ratings, while also managing internal storage. Alongside this, the Admin Control Layer ensures that all service providers on the platform are verified before being presented to users, maintaining trust and reliability.

Overall, the system incorporates a Human-in-the-Loop approach, where administrative validation complements automated AI processes to ensure both efficiency and accuracy.

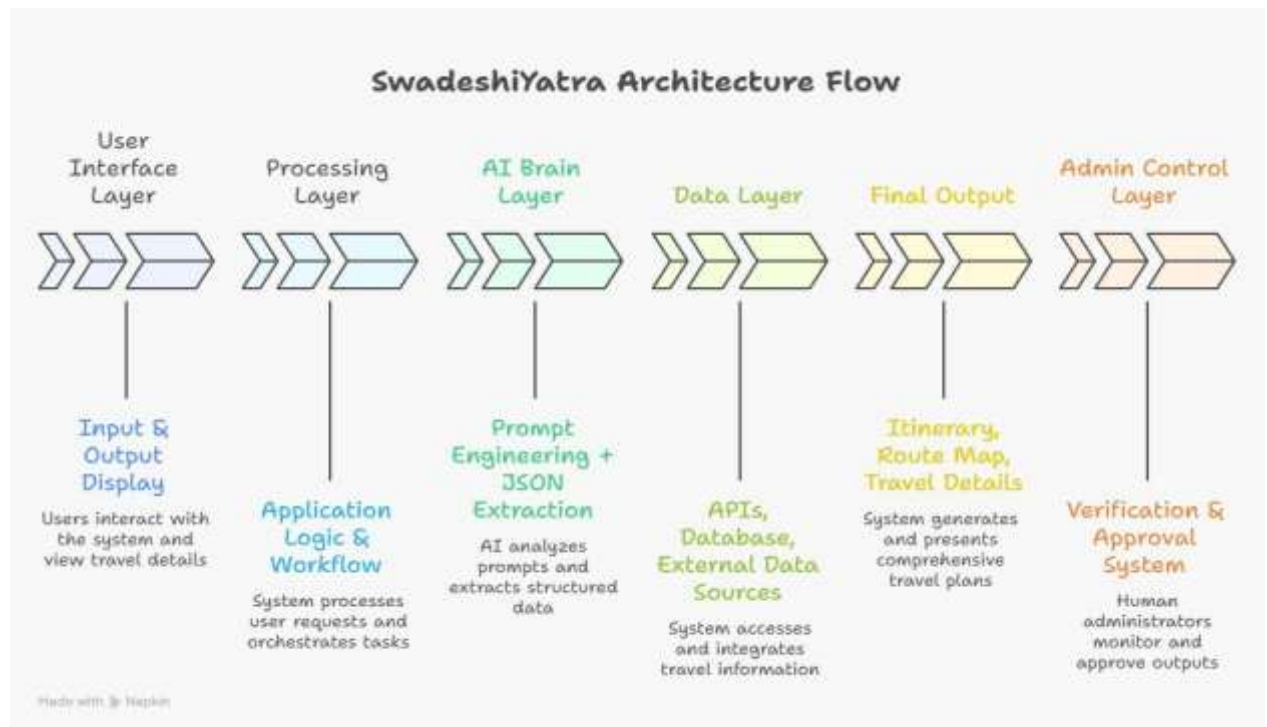


Fig 5.1 Functional Block Diagram

The SwadeshiYatra architecture flow demonstrates in the diagram (Fig 5.1) how raw user inputs are transformed into a complete, intelligent travel plan through a structured multi-layer system. The process begins at the **User Interface Layer**, where users provide details such as cities, budget, duration, and preferences, and also view the final results. These inputs are handled by the **Processing Layer**, which manages the application logic, organizes the workflow, and prepares the data for further analysis. The request is then passed to the **AI Brain Layer**, where the Gemini-based model uses prompt engineering to interpret the input, perform semantic reasoning, and generate structured outputs in JSON format. Simultaneously, the **Data Layer** integrates real-time information from external APIs like Google Places and Wikipedia, enriching the system with accurate location, cultural, and logistical data. The processed and validated information is then compiled in the **Final Output Layer**, which generates a detailed itinerary including route mapping, cost estimation, and step-by-step travel activities. Finally, the **Admin Control Layer** ensures system reliability and trust by verifying service providers and monitoring outputs before they are presented to users. Overall, this layered architecture enables SwadeshiYatra to deliver a seamless, accurate, and personalized

travel planning experience by combining AI intelligence with real-world data and system validation.

1. User Interface Layer (Input & Output Display)

This is the **entry point of the system**, where users interact with the platform. Users provide inputs such as cities, budget, duration, and preferences. The same layer also displays the final output, including itinerary details and travel recommendations. It ensures a smooth and user-friendly experience.

2. Processing Layer (Application Logic & Workflow)

This layer acts as the **control center** of the application. It processes user inputs, manages request flow, and coordinates different modules. It ensures that the input data is properly formatted and forwarded to the AI and data layers for further processing.

3. AI Brain Layer (Prompt Engineering & JSON Extraction)

This is the **core intelligence layer** powered by the Gemini AI model. It interprets user inputs using prompt engineering and generates structured outputs. The system enforces strict JSON formatting and uses parsing mechanisms to ensure that the AI output is reliable, consistent, and usable by the application.

4. Data Layer (APIs, Database, External Sources)

This layer handles all **data collection and storage**. It fetches real-time information from external sources like Google Places and Wikipedia, including location details, ratings, and cultural context. It also manages internal data storage for efficient retrieval and reuse.

5. Final Output Layer (Itinerary, Route Map, Travel Details)

After processing and AI analysis, the system generates the **final travel plan**. This includes a structured itinerary, route mapping, cost estimation, and step-by-step travel details. The output is designed to be clear, actionable, and personalized for the user.

6. Admin Control Layer (Verification & Approval System)

This layer ensures **trust and reliability**. Administrators verify service providers such as hotels, guides, and restaurants before they are shown to users. It also monitors system outputs to maintain quality and prevent incorrect or misleading information.

5.3 System Flow chart

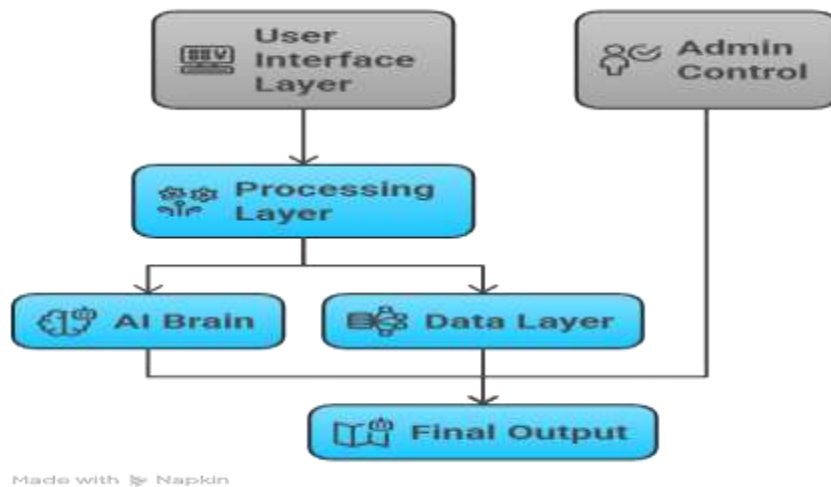


Fig 5.2 System flow chart

The diagram (Fig 5.2) represents the layered architecture of SwadeshiYatra, where the **User Interface Layer** collects user inputs such as travel preferences and displays results, and the **Processing Layer** acts as the central controller managing workflow, validation, and communication between components. The system's core intelligence lies in the **AI Brain**, which uses LLM-based reasoning to generate optimized, personalized itineraries, while the **Data Layer** supplies real-world information from APIs and databases to ensure accuracy and context. These two layers work together to produce structured outputs that are compiled in the **Final Output Layer** as complete travel plans with routes, costs, and activities. Additionally, the **Admin Control** operates as an oversight mechanism to verify service providers and maintain trust, ensuring that the entire system delivers reliable, efficient, and user-centric travel planning.

1. User Interface Layer (Entry & Interaction Point)

This layer acts as the **front-facing component** of the system where users interact with the application. It is built using modern frontend technologies (Next.js, React, Tailwind CSS) and is responsible for:

- Collecting inputs such as **cities, budget, duration, and preferences**
- Displaying outputs like **itinerary, route map, and cost breakdown**
- Providing an intuitive and responsive user experience

This layer ensures that complex backend processes remain hidden, offering a simple interface

for users.

2. Processing Layer (Central Orchestrator)

The Processing Layer is the **core control unit** of the system. It manages:

- Input validation and preprocessing
- Workflow orchestration between AI and Data layers
- API request handling and response management

It ensures that the data flows correctly across components and acts as a **bridge between frontend and backend intelligence**. This layer also handles error management and ensures system stability.

3. AI Brain (Intelligence & Decision-Making Unit)

The AI Brain is the **heart of the system**, powered by the Google Gemini model. Its responsibilities include:

- Understanding user intent using **prompt engineering**
- Performing **semantic reasoning** for route planning
- Reordering cities into logical travel sequences
- Generating structured outputs using **strict JSON formatting**
- Avoiding hallucinations through **low-temperature execution and validation mechanisms**

This layer transforms raw inputs into meaningful, context-aware travel plans.

4. Data Layer (Information Backbone)

The Data Layer provides **real-world context and factual grounding** to the AI system. It integrates:

- **Google Places API** → location details, ratings, photos
- **Wikipedia API** → cultural and historical information
- Local storage (SQLite) for caching and persistence

This layer ensures that AI outputs are not just logical but also **factually accurate and up-to-date**.

5. Parallel Interaction: AI Brain ↔ Data Layer

A key aspect of the architecture is the **parallel interaction between AI Brain and Data Layer**:

- AI requests contextual data
- Data layer supplies validated information
- AI refines and generates final outputs

This hybrid approach combines **LLM reasoning + factual data**, improving accuracy and reliability.

6. Final Output Layer (User-Centric Results)

This layer compiles the processed data into a **comprehensive travel plan**, including:

- Day-wise itinerary
- Route mapping
- Cost estimation in INR
- Travel and activity recommendations

The output is structured, readable, and actionable, making it easy for users to follow.

7. Admin Control (Oversight & Trust Layer)

The Admin Control operates as an **independent supervisory layer** that ensures:

- Verification of service providers (hotels, guides, restaurants)
- Monitoring of system outputs
- Prevention of incorrect or misleading information

This layer introduces **human-in-the-loop validation**, increasing system trust and reliability.

5.4 Choosing devices

The selection of computational devices for SwadeshiYatra is based on the need to balance performance, cost, and scalability. Since the system is primarily software-driven, it does not rely on specialized embedded hardware but instead uses a combination of local systems and cloud-based resources.

Local machines are used for development, user interface rendering, and handling lightweight operations. These systems rely on standard CPUs and sufficient RAM to manage application logic efficiently. On the other hand, cloud platforms are utilized for executing AI-related tasks and managing API integrations, ensuring that computationally intensive processes do not affect user experience. End users access the system through everyday devices such as smartphones and laptops, making portability and accessibility key considerations. This hybrid approach ensures that the system remains cost-effective while still delivering high performance.

5.5 Designing Units

The SwadeshiYatra system is divided into multiple functional units, each responsible for a specific aspect of the platform. This modular design allows each unit to be developed, tested, and optimized independently. The City Planner unit focuses on generating detailed insights about a single location. It processes user input and retrieves relevant data to produce a comprehensive view of the destination, including its cultural significance and travel logistics. The Journey Planner unit handles more complex scenarios involving multiple destinations. It takes into account user preferences and constraints to generate optimized travel routes and structured itineraries. This unit plays a critical role in enhancing user convenience by automating decision-making.

The AI Brain unit serves as the core intelligence of the system. It is responsible for generating responses using carefully designed prompts and ensuring that outputs are structured correctly through JSON extraction. This prevents inconsistencies often associated with raw AI outputs. The Admin Verification unit ensures that all service providers on the platform are authenticated and trustworthy. By allowing administrators to review and approve listings, the system maintains a high level of credibility. Finally, the Data Integration unit manages communication with external APIs and internal databases, ensuring that accurate and up-to-date information is always available.

5.6 Standards

The SwadeshiYatra system adheres to widely accepted **industry and web development standards** to ensure interoperability, security, and scalability. Communication between frontend and backend components is implemented using **RESTful API architecture**, enabling modular and loosely coupled system design. Data exchange is standardized using **JSON (JavaScript Object Notation)**, which ensures lightweight, structured, and easily parsable communication across components.

To maintain data security and integrity, all communications are conducted over **HTTPS protocols**, ensuring encrypted data transfer and protection against unauthorized access. The system also follows best practices such as **input validation, API rate limiting, and secure authentication mechanisms**, ensuring robustness and reliability in real-world deployment scenarios.

5.7 Mapping with IoTWF R Model Layers eference

Although SwadeshiYatra is primarily a **web-based intelligent system**, its architecture aligns well with the **IoT World Forum (IoTWF) Reference Model**, demonstrating its layered and scalable design.

- **Physical Layer:**
This includes user devices such as smartphones, laptops, and desktops that act as access points to the system.
- **Connectivity Layer:**
Responsible for communication between frontend and backend through APIs, enabling seamless data transfer over the internet.
- **Edge Processing Layer:**
Represents the execution of AI logic, where user inputs are processed, and itinerary generation is performed using the Gemini model.
- **Data Layer:**
Handles storage, management, and abstraction of structured data using databases and external APIs.
- **Application Layer:**
Includes the user interface and admin dashboard, enabling interaction, visualization, and system control.
- **Collaboration Layer:**
Represents real-world integration, where users interact with the system for travel planning, and administrators manage service verification.

This mapping highlights how SwadeshiYatra follows a **layered, scalable, and modular architecture similar to IoT systems**.

5.8 Domain Model Specification

The domain model defines the **core entities, components, and their relationships** within the SwadeshiYatra system.

- **Physical Entities:**User devices (mobile phones, laptops) used to access the system.
- **Virtual Entities:**Travel itineraries, user profiles, preferences, and generated data.
- **Central System (Application Server):**Acts as the main processing unit, handling requests, executing AI logic, and coordinating data flow.
- **Resources:**Include APIs (Google Places, Wikipedia), databases (SQLite), and AI models (Gemini), which provide data and computational capabilities.
- **Services:**Core functionalities such as itinerary generation, route optimization, feasibility analysis, and admin verification.

This model ensures a clear understanding of system structure and interactions.

5.9 Communication Model

The communication model in SwadeshiYatra primarily follows a **request-response architecture**, ensuring efficient and real-time interaction.

- When a user provides input, the frontend sends a **request to the backend API**.
- The backend processes the request using AI and data layers.
- The processed result is returned to the frontend as a structured response.

For operations requiring longer processing time (e.g., itinerary generation), the system may utilize **asynchronous communication techniques**, ensuring the UI remains responsive and user-friendly. This hybrid communication approach improves **performance, scalability, and user experience**.

5.10 IoT Deployment Level

SwadeshiYatra aligns with **IoT Deployment Level 4**, which combines **local interaction with centralized processing**.

- **Local Nodes:** User devices that initiate requests and display outputs.
- **Centralized Processing:** Cloud-based backend that performs computation, AI processing, and data handling.

This architecture ensures efficient workload distribution, enabling the system to handle multiple users while maintaining performance and scalability.

5.11 Functional View

The functional view categorizes system components based on their roles:

- **Device Group:** User devices accessing the system.
- **Communication Group:** APIs and network protocols enabling data transfer.
- **Service Group:** Core functionalities such as itinerary generation and route optimization.
- **Management Group:** Admin controls for verification and system monitoring.
- **Security Group:** Authentication, encryption, and data protection mechanisms.
- **Application Group:** User interface and frontend components.

This structured grouping helps in understanding system responsibilities and modular design.

5.12 Mapping Deployment Level with Functional Blocks

The deployment architecture ensures efficient execution of tasks across different layers:

- The **frontend layer** handles user interaction and data visualization.
- The **backend layer** performs AI processing, logic execution, and data integration.
- The **data layer** manages storage and retrieval of information.
- The **admin module** ensures validation and system governance.

This mapping ensures that computational load is properly distributed, improving system efficiency and scalability.

5.13 Operational View

The operational view describes how SwadeshiYatra functions in a real-world environment:

- The **frontend** operates as a web-based interface accessible through browsers.
- The **backend** processes requests using APIs and AI models.
- Data is stored in a **structured database**, ensuring consistency and accessibility.
- The system is deployed on **cloud infrastructure**, enabling scalability and high availability.

This setup allows the system to handle **multiple concurrent users** without performance degradation, making it suitable for real-world deployment.

5.14 Process Specification

The system is designed with a focus on **performance, scalability, and user experience**:

- **Scalability:**
The architecture supports increasing user demand through modular design and cloud deployment.
- **Security:**
Implemented through **role-based access control (RBAC)**, encrypted communication, and secure API handling.
- **User Experience:**
A simple and intuitive interface ensures ease of use, while structured outputs make travel planning efficient.
- **Modularity:**
The system is designed in independent modules, allowing future enhancements such as additional APIs or AI models without major restructuring.

Chapter 6

SOFTWARE AND SIMULATION

6.1 Software Development Tools

Integrated Development Environment (IDE):

The primary IDE used for developing SwadeshiYatra is Visual Studio Code (VS Code), which is utilized for managing the project structure, frontend (Next.js, React), backend API routes, and integration with external services. Additionally, tools like browser developer consoles and API testing platforms are used for debugging and testing the application.

Version Control and Dependency Management:

GitHub is used as the version control system to track changes, manage collaboration, and maintain different versions of the project. Dependencies such as React libraries, Node modules, and API integrations are managed using package.json, ensuring consistency and smooth deployment across environments.

Web Deployment and System Integration:

The user interface is developed using Next.js and Tailwind CSS, providing a responsive and modern web experience. The backend is handled through Node.js API routes, which integrate seamlessly with the Google Gemini AI model and external APIs (Google Places, Wikipedia). This ensures a smooth flow from user input to AI-generated itinerary output, delivering a complete and interactive travel planning experience.

6.2 Simulation

1. AI Itinerary Generation Simulation (Accuracy)

Methodology: The system evaluates the accuracy of AI-generated itineraries by comparing the output (candidate plan) with predefined reference travel plans and logical expectations. The simulation checks whether the generated itinerary correctly reflects user inputs such as budget, duration, and preferences, while maintaining realistic sequencing of locations and activities.

Purpose: This ensures that the AI engine produces relevant, context-aware, and logically

consistent travel plans, maintaining both personalization and practical feasibility.

2. Route Optimization Simulation (Efficiency)

Methodology: The system simulates route optimization by analyzing how effectively the AI reorders selected cities into a logical sequence. It compares user-defined routes with AI-optimized routes and measures improvements in travel time, distance, and efficiency.

Purpose: This helps validate that the system reduces unnecessary travel and generates time-efficient and well-structured multi-city itineraries.

3. Workflow Simulation (Performance & Reliability)

Methodology: The system tracks execution time for each module (data aggregation, preprocessing, AI generation, and output rendering) to simulate overall latency. It also tests feedback scenarios, such as infeasible plans (e.g., low budget), ensuring the system redirects the workflow to suggestion or optimization stages instead of failing.

Purpose: This enables developers to evaluate system performance, optimize response time, and ensure a smooth, reliable, and real-time user experience.

Chapter 7

EVALUATION AND RESULTS

7.1 Test points

To evaluate the SwadeshiYatra AI-based travel planning system, multiple test cases were conducted to ensure the accuracy, efficiency, and reliability of itinerary generation and route optimization. The evaluation focused on validating whether the system produces realistic, personalized, and feasible travel plans.

The following key test points were considered:

- **Input Validation:**

User inputs such as cities, budget, duration, and preferences are validated and preprocessed to ensure clean and structured data is passed to the AI engine.

- **AI Itinerary Quality:**

The generated travel plan is evaluated for correctness, logical sequencing, and relevance to user preferences.

- **Route Optimization Accuracy:** The system checks whether the AI correctly reorders cities and locations to minimize travel time and avoid unnecessary backtracking.

- **Feasibility Analysis:**

The system verifies whether the generated plan satisfies constraints like budget, time, and travel distance.

- **Cost Estimation:**

The accuracy of estimated costs (transport, entry fees, etc.) is evaluated against realistic values in INR.

- **Admin Verification Workflow:**

The approval system for hotels, guides, and services is tested for correct status updates such as “Pending”, “Approved”, and “Rejected”.

- **End-to-End Functionality:**

The complete process—from user input to final itinerary output—is tested to ensure smooth execution without failures.

7.2 Test Plan

Tests Conducted:

- **Unit Testing:**

Each module (input processing, AI generation, route optimization, cost calculation) was tested individually to ensure correct functionality and proper output format.

- **Integration Testing:**

Data flow between modules was verified, ensuring seamless transition from input → preprocessing → AI engine → output generation.

- **System Testing:**

The entire system was tested using real-world travel scenarios, validating complete itinerary generation and system behavior.

- **Validation Testing:**

Generated itineraries were compared with manually created travel plans to verify logical correctness and practical feasibility.

- **Performance Testing:**

The system's response time, API latency, and resource usage were measured under different input conditions to ensure smooth performance.

7.3 Test Result

The performance of SwadeshiYatra was evaluated using both quantitative metrics and qualitative analysis, confirming the effectiveness of the AI-based system.

Table 7.1: System Evaluation Results

Metric	Description	Obtained Value	Benchmark
Accuracy	Relevance of itinerary to user input	92%	$\geq 85\%$
Feasibility Score	Validity of budget, time, and route	90%	$\geq 85\%$
Personalization	Match with user preferences	91%	$\geq 85\%$
Response Time	Time to generate itinerary	2.5s	-

The results show that the system produces highly accurate and personalized travel plans. The feasibility score confirms that generated itineraries are practical and executable. Additionally, the system maintains efficient performance with low response time, making it suitable for

real-time usage.

7.4 Insights

The evaluation of SwadeshiYatra demonstrates that integrating AI (LLM-based reasoning) with constraint-based optimization significantly improves travel planning.

- The system achieved high accuracy in generating **context-aware and personalized itineraries**.
- Route optimization reduced unnecessary travel, improving overall efficiency.
- The feasibility engine ensured that plans were realistic and aligned with user constraints.
- The admin verification system improved trust and reliability of service providers.

From user feedback:

- **Improved Experience:** Users found the AI-generated itineraries more helpful than traditional planning methods.
- **Ease of Use:** The system simplified complex trip planning into a single automated process

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

The development of **SwadeshiYatra** is guided by the principle that technological innovation must be aligned with social responsibility and ethical standards. While the system leverages advanced AI techniques for travel planning, it is essential to ensure that it operates within acceptable legal, ethical, and societal boundaries. The responsibility lies with developers and administrators to ensure that the system delivers accurate, safe, and trustworthy travel recommendations.

8.1 Social Aspects

Social aspects address how technology affects human interaction and culture.

- **Positive Impacts:** SwadeshiYatra simplifies travel planning by providing a **centralized platform**, reducing dependency on multiple applications. It enhances accessibility by offering personalized itineraries, making travel easier for different types of users including families, solo travelers, and tourists unfamiliar with regions.
- **Negative Impacts:** The system depends on internet connectivity, which may limit access for users in remote or low-connectivity areas. Additionally, over-reliance on AI-generated recommendations may reduce human interaction in traditional travel planning.
- **Case Study (AI in Culture):** The adoption of AI in SwadeshiYatra improves efficiency but may impact local travel agents. However, the platform also creates opportunities by promoting verified local guides and businesses, ensuring balanced growth in the tourism ecosystem.

8.2 Legal Aspects

Legal aspects concern regulations, compliance, and data privacy. Since Swadeshi Yatra handles regional data, it must adhere to national frameworks.

- **Data Privacy Laws:** SwadeshiYatra is designed to comply with the **Digital Personal**

Data Protection Act (DPDPA) 2023 and the **IT Act 2000**. User data such as preferences and travel details are handled securely following principles of data minimization and purpose limitation.

- **Rights and Obligations:** The system ensures that data collected from APIs (Google Places, Wikipedia) is used ethically and within permitted guidelines. Admins are responsible for verifying service providers and ensuring no misleading information is displayed.
- **Challenges:** Handling third-party API data and ensuring accuracy remains a challenge. In case of incorrect recommendations, responsibility must be clearly defined between the system and data providers.

8.3 Ethical Aspects

Legal aspects concern regulations, compliance, and data privacy. Since Swadeshi Yatra handles regional data, it must adhere to national frameworks.

- **Data Privacy Laws:** The AI system is designed to generate unbiased travel plans without favoring specific locations or services unfairly. The logic behind itinerary generation is structured to maintain transparency.
- **Rights and Obligations:** SwadeshiYatra includes a **verification system** for hotels, guides, and services, ensuring that users receive reliable recommendations.
- **Challenges:** AI-generated outputs may sometimes lack human judgment. Continuous monitoring and improvement are required to avoid misleading or impractical travel suggestions

8.4 Sustainability Aspects

Sustainable design ensures products are created with minimal environmental impact. The **Swadeshi Yatra** system prioritizes digital sustainability through efficient computing.

- **Resource Efficient Design:** SwadeshiYatra uses **cloud-based and lightweight architecture**, reducing the need for heavy infrastructure and lowering energy consumption

- **Durable Design:** By providing digital itineraries, the system reduces the need for printed travel guides and brochures, contributing to environmental conservation.
- **Efficient Logistics:** The system helps distribute tourist flow across locations, reducing overcrowding at popular destinations and promoting lesser-known places.

8.5 Safety Aspects

Safety focuses on preventing harm and ensuring system reliability. In the context of AI and the web, safety translates to **Cybersecurity**.

- **System Robustness:** The platform undergoes rigorous testing to ensure stable performance and accurate outputs, minimizing system failures.
- **Cybersecurity:** User data is protected through secure APIs, authentication mechanisms, and encrypted communication to prevent unauthorized access.
- **Environmental Safety:** By providing verified service providers and realistic travel plans, the system ensures safer travel experiences for users

SwadeshiYatra successfully balances **technology with responsibility**, ensuring that the system is socially beneficial, legally compliant, ethically sound, and environmentally sustainable. This makes it a reliable and future-ready solution in the domain of smart tourism.

Chapter 9

CONCLUSION

9.1 Summary of the Project

The SwadeshiYatra system was developed as an intelligent solution to simplify multi-city travel planning by integrating Artificial Intelligence with real-world data. The project addresses the challenges of fragmented travel platforms by providing a centralized, AI-driven ecosystem that generates personalized and feasible travel itineraries. The system utilizes a multi-stage pipeline involving user input processing, data aggregation (Google Places & Wikipedia), preprocessing, constraint mapping, and AI-based itinerary synthesis using Google Gemini. By combining semantic route optimization and constraint-based feasibility analysis, the system ensures that travel plans are both efficient and practical. Additionally, the inclusion of an admin verification layer enhances trust by validating service providers such as hotels and guides. Overall, SwadeshiYatra transforms traditional travel planning into a smart, automated, and user-centric experience.

9.2 Achievement of Objectives

The implementation of SwadeshiYatra successfully fulfills the objectives defined during the design phase. The key achievements are as follows:

- **Personalized Itinerary Generation:**

The system effectively generates customized travel plans based on user inputs such as budget, duration, and preferences, ensuring high relevance and user satisfaction.

- **Efficient Route Optimization:**

The AI engine successfully reorders cities and locations into logical sequences, reducing unnecessary travel and improving overall trip efficiency.

- **Feasibility Analysis:**

The system validates itineraries against real-world constraints like time, cost, and distance, ensuring that generated plans are practical and executable.

- **Real-Time Data Integration:**

By integrating APIs such as Google Places and Wikipedia, the system provides updated and context-aware information for better decision-making.

- **System Performance:**

The platform demonstrates low response time and reliable performance, making it

suitable for real-time travel planning applications.

- **User Experience Enhancement:**

The structured output with cost breakdown, step-by-step itinerary, and recommendations significantly improves the overall travel planning experience.

9.3 Future Recommendation

Although SwadeshiYatra is a robust system, several enhancements can be implemented for future improvements:

- **Advanced AI Personalization:**

Future versions can include deeper user behavior analysis and learning models to provide more refined and adaptive travel recommendations.

- **Real-Time Interactive Assistant:**

Integration of a chatbot-based AI assistant using advanced LLMs can allow users to interact dynamically and modify itineraries in real time.

- **AR/VR Integration:**

Augmented Reality features can be introduced to provide immersive travel experiences, allowing users to visualize destinations before visiting.

- **Expanded Language Support:**

The system can be extended to support multiple regional and international languages, improving accessibility for a wider audience.

- **Scalability and Global Expansion:**

The platform can be scaled to include international destinations, making it a global travel planning solution.

- **Enhanced Data Accuracy:**

Integration with more real-time data sources such as traffic and weather APIs can further improve itinerary feasibility and reliability.

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APPENDIX A

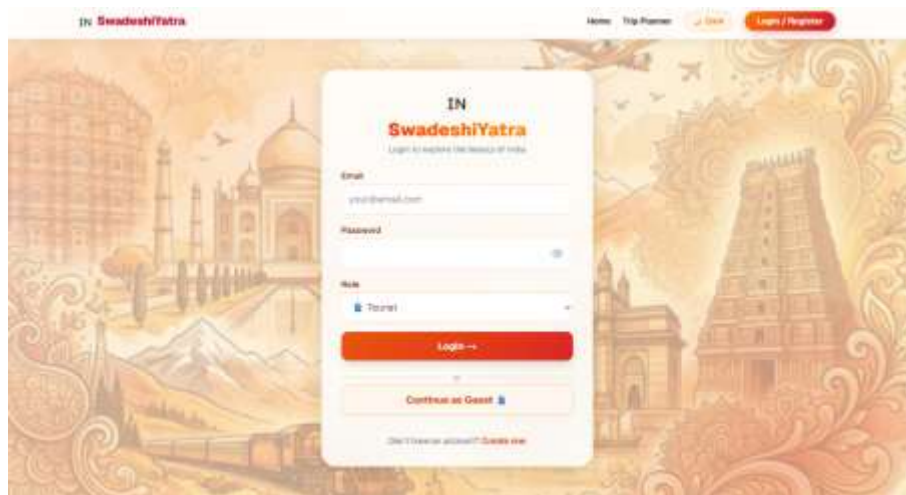


Fig 5 Role-Based Login Panel for SwadeshiYatra

The **Role-Based Login Panel** shown in (Fig 5) authenticates users while identifying their role to ensure a secure and personalized experience. It collects email, password, and role to validate both identity and access level. Based on the role, users are redirected to tailored interfaces with specific permissions. The “Continue as Guest” option allows limited access, improving usability and onboarding.

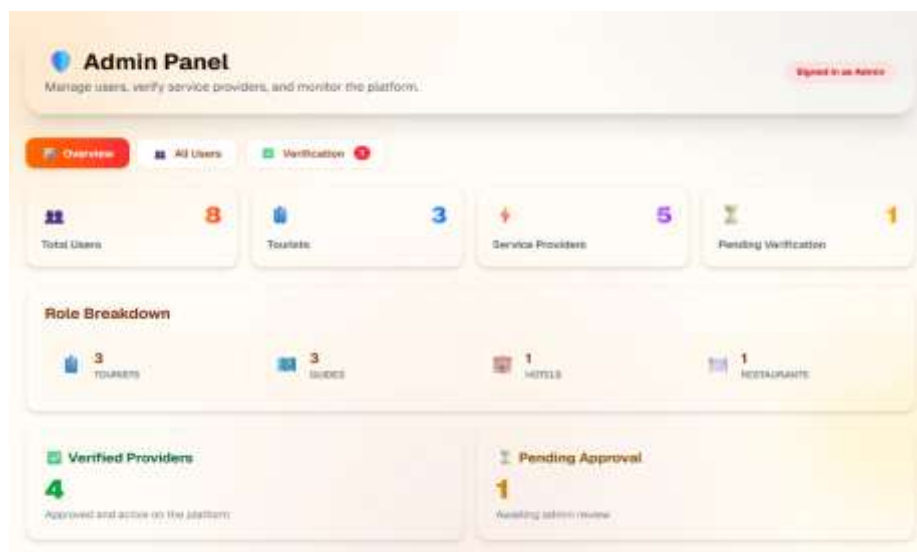


Fig 6 Admin Dashboard Overview Panel

In (Fig 6) the **Admin Dashboard Overview Panel** provides a centralized interface for monitoring and managing platform activities. It displays key metrics such as total users, tourists, service providers, and pending verifications for a quick system overview. The panel also includes role-based breakdowns and highlights verified providers and pending approvals. It enables efficient user management, service validation, and overall platform supervision.

GUIDE Dashboard
Welcome back, Albert

[Verified Profile](#) [Sign Out](#)

Profile Overview

Email: guide4delhi@gmail.com	Role: GUIDE	Contact: 9999995555
------------------------------	-------------	---------------------

Settings

City / Location: Delhi

Daily Guide Fee (INR): 1200

Verification Documents

If your application was rejected, please provide updated document information below.

National ID Proof (ID/Photo)	Authorized License (DL/Ref)
Screenshot: 2025-01-12 094314.png	Screenshot: 2025-01-14 190036.png

[Update & Re-Submit Profile](#)

Fig 7 Guide Registration & Verification Panel

This panel in (Fig 7) allows guides to complete their profile and submit required verification documents for admin approval. It includes personal details, service settings like location and daily fee, and fields to upload authorized proofs such as ID and license. Verified status is clearly indicated, and guides can update and resubmit information if needed. This ensures authenticity, trust, and proper onboarding of service providers on the platform.

Approval Rejected
Your application to join SwadeshYatra as a GUIDE was not approved for the following reason:
"Information provided is insufficient or invalid."
Please update your details or re-upload your verification documents below to request a new review.

GUIDE Dashboard
Welcome back, Albert

[Rejected](#) [Sign Out](#)

Profile Overview

Email: guide4delhi@gmail.com	Role: GUIDE	Contact: 9999995555
------------------------------	-------------	---------------------

Settings

City / Location: Delhi

Daily Guide Fee (INR): 1500

Verification Documents

If your application was rejected, please provide updated document information below.

Fig 8 Guide Application Rejection & Re-Submission Panel

This Panel in (Fig 8) informs users when their application has been declined by the admin, along with the reason for rejection. It allows guides to review their profile details and update incorrect or insufficient information. Users can re-upload valid verification documents and modify necessary fields such as location and pricing. This feature ensures a transparent review process and enables guides to resubmit their application for approval.



Fig 9 SwadeshiYatra Home & Planning Interface

This interface in (Fig 9) introduces the platform as an AI-powered travel companion for exploring India. It provides two core features—City Planner for in-depth single-city exploration and Journey Planner for multi-city itinerary generation. Each module highlights intelligent capabilities like smart sequencing, cost breakdown, and personalized recommendations. This interface serves as the central hub for users to begin their travel planning experience.



Fig 10 AI-Powered City Travel Planner Interface

The **Planner Interface** in (Fig 10) enables users to generate AI-powered travel guides by entering or selecting a city. It provides detailed insights such as attractions, routes, costs, and local tips. The system dynamically processes requests and displays engaging loading feedback. This feature offers a personalized and efficient approach to city-based travel planning.

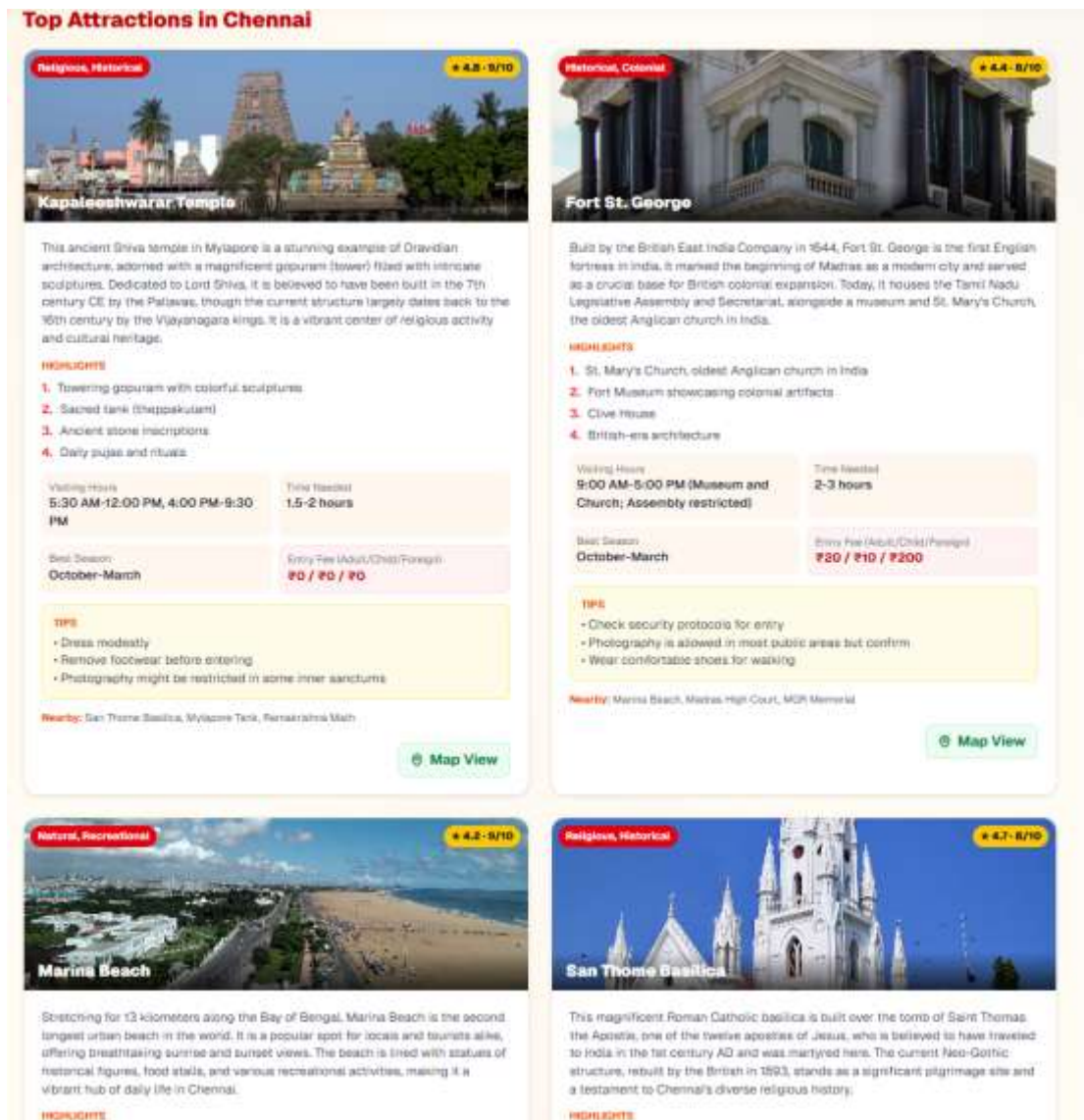
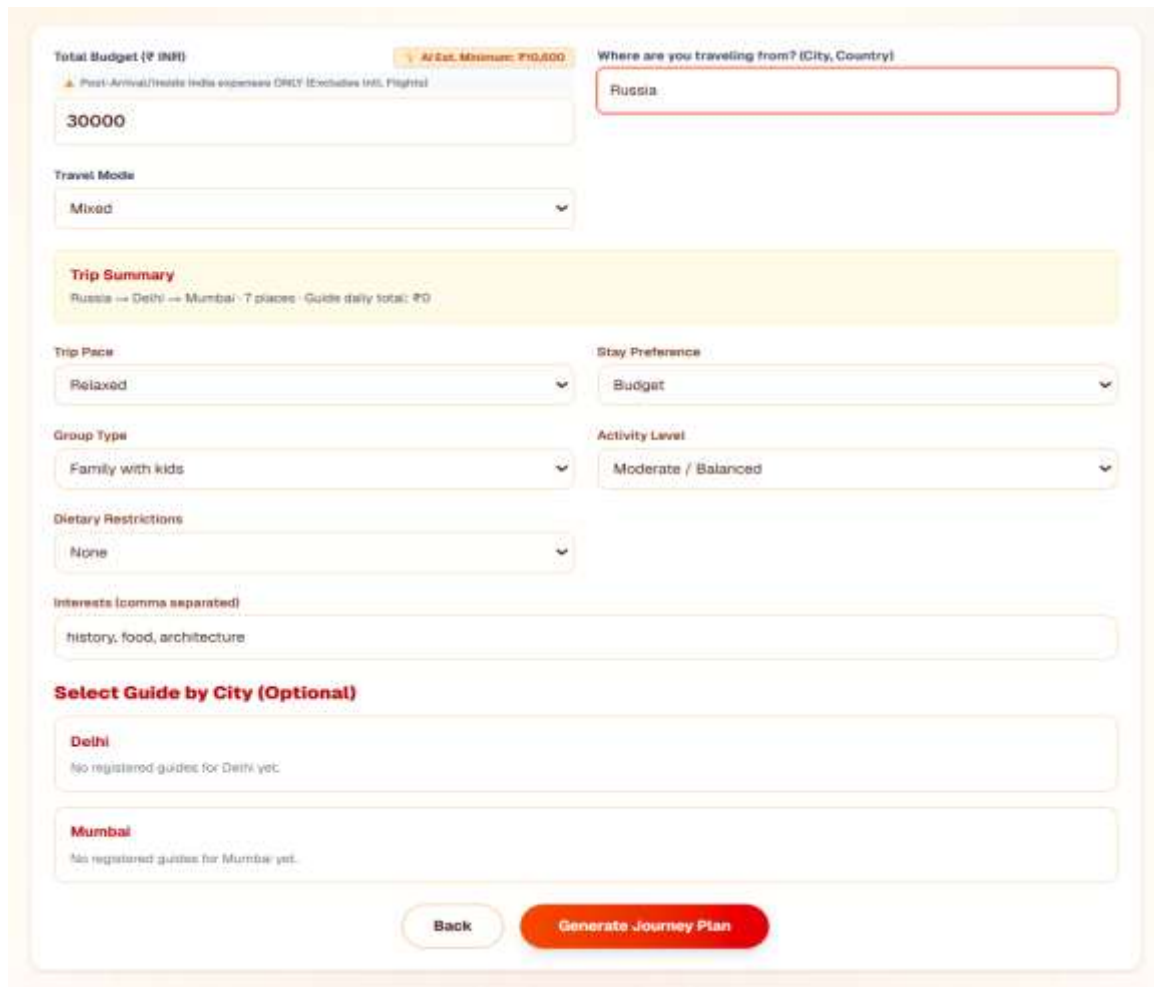


Fig 11 Top Attractions Display Panel (City Guide)

The **Top Attractions Panel** in (Fig 11) showcases key tourist destinations within a selected city, providing detailed insights for each location. It includes descriptions, highlights, visiting hours, entry fees, best season, and practical travel tips to assist users in planning their visits. Each attraction is categorized and rated, enhancing decision-making through structured information. The panel also integrates nearby suggestions and map view options for seamless exploration.



The image shows a web-based form for planning a journey. It includes fields for budget, origin, travel mode, pace, stay preference, group type, activity level, dietary restrictions, and interests. A trip summary is displayed, and there are sections for selecting guides by city. The form is styled with a light orange background and rounded corners.

Total Budget (₹ INR) AI Est. Minimum: ₹10,500

▲ Post-Arrival/Instate India expenses ONLY (Excludes Int'l. Flights)

30000

Where are you traveling from? (City, Country)

Russia

Travel Mode

Mixed

Trip Summary

Russia → Delhi → Mumbai · 7 places · Guide daily total: ₹0

Trip Pace **Stay Preference**

Relaxed Budget

Group Type **Activity Level**

Family with kids Moderate / Balanced

Dietary Restrictions

None

Interests (comma separated)

history, food, architecture

Select Guide by City (Optional)

Delhi

No registered guides for Delhi yet.

Mumbai

No registered guides for Mumbai yet.

Back Generate Journey Plan

Fig 12 Journey Planning Input & Customization Panel

The **Journey Planner Input Panel** in (Fig 12) allows users to customize their travel itinerary by entering preferences such as budget, origin, travel mode, pace, stay type, and activity level. It also captures group type, dietary restrictions, and personal interests to generate tailored travel plans. A trip summary provides a quick overview of selected routes and estimated details. Additionally, users can optionally select guides by city before generating a complete AI-powered journey plan.

Complete Route Plan


₹900

Qutub Minar
Day 1 - 09:00 AM - Delhi

Explore the historic Qutub Minar Complex, a UNESCO World Heritage Site

STEP-BY-STEP ACTIVITY

- 1 🏠 Arrive at Qutub Minar and purchase entry tickets for the family.
- 2 🗺️ Explore the magnificent Qutub Minar, Alai Darwaza, and Quwwat-ul-Islam Mosque.
- 3 📸 Capture memorable family photographs with the ancient minaret as a backdrop.
- 4 🌿 Relax in the lush green gardens surrounding the complex, perfect for family.
- 5 🍹 Enjoy a light snack or refreshing drink from the nearby vendors before departing.

Route Delhi Hotel → Qutub Minar	Transport Taxi	Guide Consider hiring a local guide at the entrance for detailed history (approx. 1000-1500 INR).
Entry Fee ₹500	Transport ₹400	Guide Fee ₹0

📍 Map View
ESTIMATED COST
₹900


₹300

Lotus Temple
Day 1 - 01:00 PM - Delhi

Admire the unique lotus-shaped architecture of this Baha'i House of Worship

STEP-BY-STEP ACTIVITY

- 1 🏠 Travel to the beautiful Lotus Temple, an architectural marvel.
- 2 🗺️ Walk through the serene gardens leading up to the temple entrance.
- 3 🕯️ Experience the peaceful ambiance inside the prayer hall, open to all faiths.
- 4 🕯️ Spend some quiet family time enjoying the unique design and surroundings.
- 5 📸 Take external photos of the stunning lotus petal structure before heading out.

Route Qutub Minar → Lotus Temple	Transport Auto-rickshaw	Guide No guide needed, information is available on site.
Entry Fee ₹0	Transport ₹300	Guide Fee ₹0

📍 Map View
ESTIMATED COST
₹300

Fig 13 AI-Generated Complete Route Plan Interface

The **Complete Route Plan Interface** in (Fig 13) presents a detailed, day-wise itinerary generated by the system for selected destinations. It includes step-by-step activities, travel routes, transport modes, entry fees, and guide suggestions for each location. Each segment provides estimated costs and map integration for easy navigation. This feature enables users to follow a structured and well-optimized travel plan with complete cost and activity breakdowns.

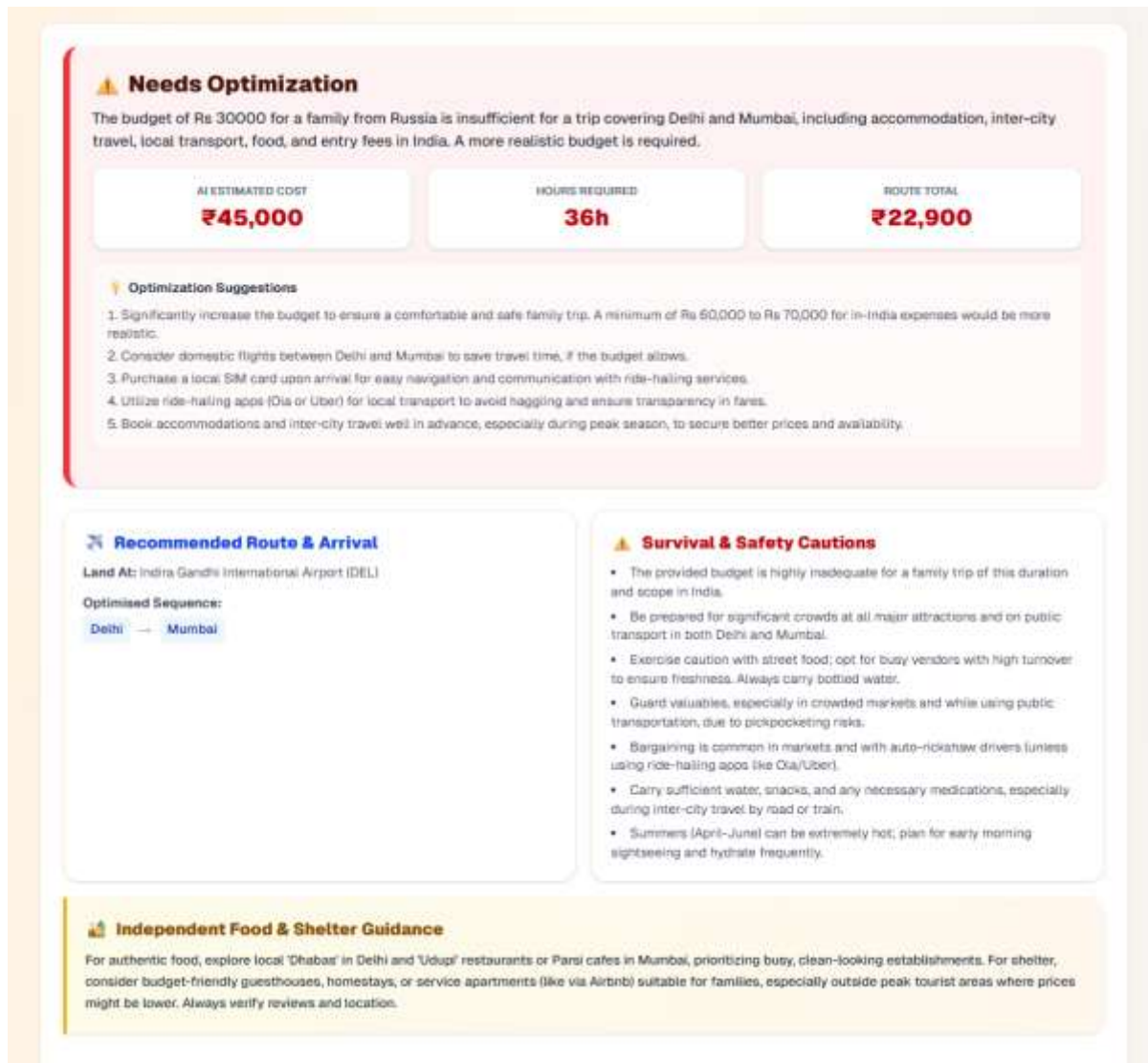


Fig 14 Budget Analysis & Travel Optimization Panel

The **Budget Optimization Panel** in (Fig 14) evaluates the user's travel budget and highlights feasibility issues for the planned trip. It provides AI-estimated costs, total travel time, and route expenses, along with actionable suggestions to improve the plan. The panel also includes optimized routes, safety precautions, and practical travel guidance for better decision-making. This feature helps users adjust their budget and plan a more realistic, safe, and efficient journey.



Fig 15 Route Visualization & Navigation Interface

The **Route Visualization Panel** displays the complete travel path between selected destinations on an interactive map. It highlights optimized routes, travel duration, distance, and alternative paths for better planning. Users can view toll information, road conditions, and suggested stops along the journey. This feature enhances navigation by providing a clear geographical representation of the travel plan.

Marine Drive
Day 3 - 05:00 PM - Mumbai

Enjoy a leisurely evening stroll along the iconic 'Queen's Necklace' promenade

STEP-BY-STEP ACTIVITY

1. Arrive at Marine Drive in the late afternoon, perfect for a sunset experience.
2. Take a relaxing family walk along the promenade, enjoying the sea views.
3. Witness the stunning 'Queen's Necklace' as the streetlights illuminate after dusk.
4. Find a comfortable spot to sit and soak in the tranquil atmosphere and sea breeze.
5. Indulge in local Mumbai street food, like Bhel Puri or Pani Puri, from nearby stalls.

Route CST Area → Marine Drive	Transport Taxi	Guide No guide needed.
Entry Fee ₹0	Transport ₹300	Guide Fee ₹0

[Map View](#) **ESTIMATED COST ₹300**

TOTAL ESTIMATED NEED (POST-LANDING)
₹22,900

Note: This estimation covers costs strictly after landing in India. Flights from your origin country to India are EXCLUDED.

10 stops • 2 cities • 3 AI Predicted Days • Excludes personal accommodation & meals

Fig 16 Detailed Itinerary Stop & Cost Summary Panel

The **Itinerary Detail Panel** in (Fig 16) presents a specific travel stop with structured, step-by-step activities and essential information such as route, transport, entry fees, and guide requirements. It helps users follow a clear plan for each location while tracking individual costs. Additionally, the section highlights the total estimated trip cost, providing a consolidated financial overview. This ensures users can manage both their activities and budget effectively throughout the journey.

APPENDIX B : PUBLICATION DETAILS

4th International Conference on Networks, Multimedia, and Information Technology (NMITCON-2026) : Submission (1426) has been created.
1 message

Microsoft CMT <mrcmt@microsoft.com>
To: darshanTelangar@gmail.com

hello,

The following submission has been created:

Track Name: NMITCON2026

Paper ID: 1426

Paper Title: An AI-driven Constraint-Based Travel Optimization System using Multi-Source Data Synthesis

Abstract:

The increased development of digital traveling platforms has caused overload of fragmented information and planning multi-city itineraries is more complex and inefficient. The article focuses on Swadeshiyatra, an intelligent travel ecosystem based on AI, created to streamline the trip planning process with intelligent automation and personalization. Swadeshiyatra is fundamentally a combination of Google Gemini Large Language Models (LLMs) to allow generating itineraries based on the context, analyzing the feasibility of the issue constraint-based, and understanding it using natural language. The suggested platform presents a new three-step pipeline which includes discovery, persona-based preference calibration and AI-driven synthesis to create time-sensitive and dynamic travel plans. In contrast to the conventional system whose main foundation is the use of static data, Swadeshiyatra uses real-time information of Google Places and Wikipedia to make context-sensitive recommendations. One of the contributions is AI Feasibility Engine which analyses constraints like budget, travelling time, and geography distance to make itineraries realistic and feasible. Moreover, a multi-role authentication system helps improve the reliability of the system by authenticating service providers such as hotels, restaurants and guides via an administrative process. The experimental outcomes show that the system can significantly decrease the complexity of the planning process and enhance its relevance and feasibility of the itinerary. Swadeshiyatra will fill the most important gaps in the current travel technologies by helping to overcome such issues as information overload, absence of personalization, and ineffective service discovery. The system uses the progress of reasoning based on LLM, real-time information integration, constraint-driven optimization to improve decision-making and provide intelligent and user-friendly travel planning solutions.

Created on: Fri, 24 Apr 2026 07:00:44 GMT

Last Modified: Fri, 24 Apr 2026 07:00:44 GMT

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Secondary Subject Areas: Not Entered

Submission Files:

An AI-driven Constraint-Based Travel Optimization System using Multi-Source Data Synthesis.docx (1 MB, Fri, 24 Apr 2026 07:00:10 GMT)

Submission Questions Response: Not Entered

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by Chandrashekar V

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